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Editor's Note

Dear Readers,

In celebration of Stuyvesant High School's 100th birthday, the Math Survey staff is bringing to you another great issue of the Math Survey Magazine - with special features! To celebrate with us are some of Stuyvesant's alumni, who are now top-notch mathematicians. Their thoughts, encouragement, and even some of their articles that they've written for Math Survey are included in this issue.

In continuing Math Survey's tradition, we also have ten perplexing math problems. You may check your answers with the solutions provided.

More students than ever are entering articles for selection this year, so you will be able to read a variety of topics from the more traditional topics such as Euler to more modern ones like Optimal Selection of a Service Center. Almost half of the authors have been Math Fair or Intel Science Talent Search winners, some of whom have included a shortened version of their research.

For students who are new to reading math articles, although most of the mathematics is above high school level, what's important to us is that you get a general idea, rather than the specifics. Try appreciating the mathematics. If you learn to appreciate it, you'll have more patience with the articles. Take as much time as you need. It took all of us a lot of time to write it, and even more time to learn it.

We had a great time putting the magazine together, and we hope you'll love it. If you're interested, you can even join us next year. Until then, enjoy yourself!

Sincerely,
Betty Luan
Editor-in-Chief

An Introduction to Euler's Formula and Topology

By Betty Luan

The Euler polyhedron formula,

$$v + f = e + 2 \quad (1)$$

,where v , f , and e are the number of vertices, facets, and edges of a convex 3-dimensional polytope (3-polytope), was on several occasions proclaimed as one of the top ten theorems of mathematics. Besides being a useful equation to classify polytopes, Euler's formula plays a large part in algebraic topology, which I will discuss later.

We can rewrite equation 1 as

$$f_0 - f_1 + f_2 = 2 \quad (2)$$

,where f_i is the i -dimensional faces of the polytope. For d dimensional polytopes, equation 2 can be generalized as

$$f_0 - f_1 + f_2 - \dots + (-1)^{(d-1)} * f_{(d-1)} = 1 + (-1)^{(d-1)} \quad (3)$$

The f -vector of a d -dimensional polytope is $(f_0, f_1, \dots, f_{(d-1)})$. One might ask "What are the possible f -vectors of convex d -polytopes?" For 2-polytopes and 3-polytopes, this question is easy to answer. For $d=2$, the f -vectors are (f_0, f_1) , where $f_0 = f_1 \geq 3$. For $d=3$, the f -vector is (f_0, f_1, f_2) , where the conditions $f_2 \leq 2f_0 - 4$ and $f_0 \leq 2f_2 - 4$ are satisfied.

eg. 2-polytopes

1. A triangle has 3 vertices (f_0) and 3 edges (f_1).
2. A hexagon has 6 vertices (f_0) and 6 edges (f_1).

3-polytopes

1. A tetrahedron has f -vector $(4, 6, 4)$, satisfying the two conditions
 $4 \leq 2(4) - 4$ and $4 \leq 2(4) - 4$
2. A bipyramid has f -vector $(6, 12, 8)$, satisfying the two conditions
 $6 \leq 2(8) - 4$ and $8 \leq 2(6) - 4$

You may try more examples.

However, the analogue question for the 4-dimensional polytope is still unsolved. Several conditions are known to be satisfied for f -vectors (f_0, f_1, f_2, f_3) , while many others are missing. One condition that we know for the f -vector to satisfy is

$$f_1 + f_2 \leq 6(f_0 + f_3) \quad (4)$$

While many questions still have yet to be answered relating to Euler's formula, Euler's formula has spurred the development of other areas in mathematics, including algebraic topology. One important question asked in algebraic topology is "How do we know whether two surfaces are homeomorphic?" We can rephrase the question to ask, "How do we know there exists a homeomorphism from surface A to surface B . (written as $A \simeq B$)?" A homeomorphism between two surfaces is a way of bending, stretching, or deforming one to make it look like the other, only allowing cuts if they are subsequently glued back in the same place.

For example, a torus, a donut-like surface, is homeomorphic to a ring. We can attain this by stretching the torus into the shape of a ring. Is a donut homeomorphic to a tennis ball?

The previous examples have relatively simple transformations. However, given a more complicated surface, how can we be sure that it is homeomorphic to another intricately twisted surface? If there is no such transformation, how can we prove it? These questions are answered using an invariant. It is a quantity that can be evaluated for a surface, with the property that if $A \simeq B$, then $I(A)=I(B)$. Equivalently, if $I(A)$ does not equal $I(B)$, then A is not homeomorphic to B . If an invariant takes different values for two surfaces, then we know that they are not homeomorphic.

How do we find an invariant? Here, Euler's formula comes into use. For a given triangulation of a surface, the invariant can be calculated from the following equation

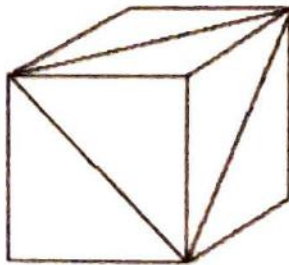
$$X = (\# \text{ of faces}) - (\# \text{ of edges}) + (\# \text{ of vertices}) \quad (5)$$

known as the Euler's characteristic.

A triangulation of a surface is a partition of the surface into triangles such that any two triangles intersect only in a vertex, an edge, or not at all. A surface may have many different triangulations, however, any two triangulations of the same surface will have the same Euler characteristic, which is why the Euler characteristic is an invariant.

Now, we are able to prove whether a donut is homeomorphic to a tennis ball by triangulating both surfaces and finding the invariants. Because triangulating a sphere is inconvenient, we may triangulate a cube instead, since a simply homeomorphism exists between the cube and the sphere. We may also use the same technique for triangulating a torus.

eg. Sphere



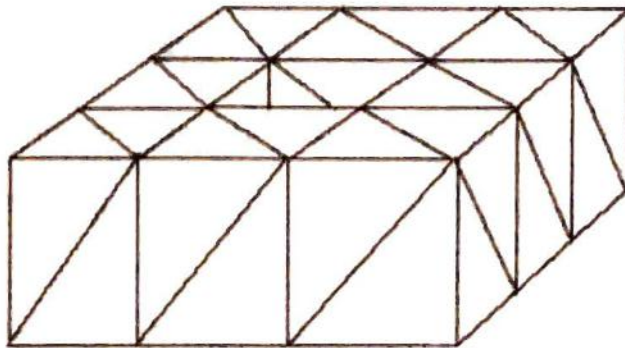
$$\#f = 12$$

$$\#e = 18$$

$$\#v = 8$$

$$X(S) = 2$$

Torus



$$\#f = 64$$

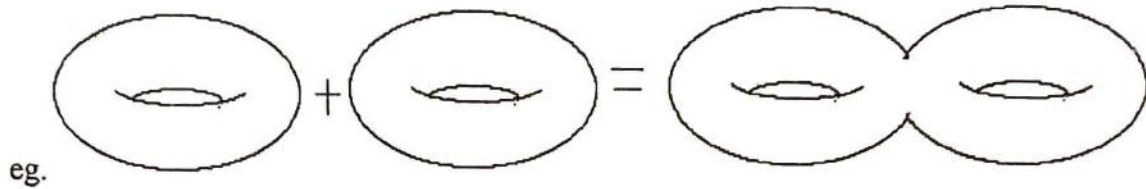
$$\#e = 100$$

$$\#v = 56$$

$$X(T) = 0$$

Because $X(S)$ does not equal $X(T)$, we know that the tennis ball is not homeomorphic to a donut.

We can use these invariants to further study connect sums of surfaces. Connect sums is the procedure of combining two surfaces to make another. First cut holes in each surfaces, then glues the surfaces together along the new boundary.

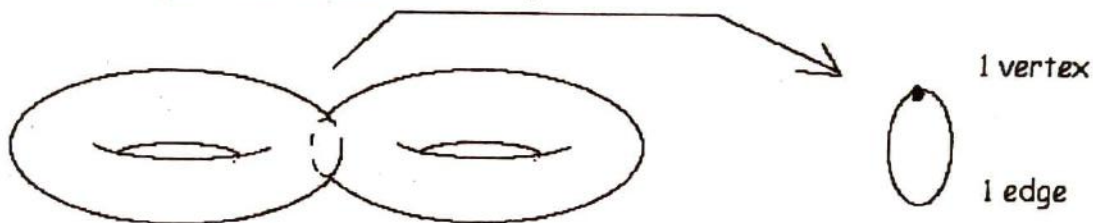


Two tori of genus 2 (having two holes) have a connect sum of genus 4. We denote the connect sum of surface A and B as $A \# B$. The new figure will have an invariant of its own, where

$$X(A \# B) = X(A) + X(B) - 2 \quad (6)$$

, where $X(A)$ and $X(B)$ are the invariants of surfaces A and B.

I will prove the formula as follows,



The net loss is 1 edge and 1 vertex, totaling 2.

We have proven the general case of any connect sum. The Euler characteristic of a connect sum of g tori can be calculated with

$$X(T_1 \# T_2) = 2 - 2g \quad (7)$$

, where g is the genus of the two tori. I will prove the formula as follows.

Proof by induction : $g = 1$ $x_0 = 2 - 2(1) = 0$

$g = n$	$x_1 = 2 - 2n$	
$g = n + 1$	$x_2 = (2 - 2n) + x_0 - 2$	using Equation 6
	$= 2 - 2n + 0 - 2$	
	$= -2n$	
	$x_2 = 2 - 2(n+1)$	using Equation 7
	$= 2 - 2n - 2$	
	$= -2n$	

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<<http://www.mathworld.org>>

A Mathematical Look at Origami

By Alice Liu

Origami, the Chinese and Japanese art of paperfolding, is fascinating as a recreational activity and a subject of mathematical studies. It is believed that in the 1st century A.D. when paper was invented in China, people had been folding it into simple designs. When the secret of paper was carried to Japan by Buddhist monks in the 6th century, it was quickly integrated into the Japanese culture. Japanese folding was then divided into ceremonious folding and recreational folding. The term "origami" was finally coined in the nineteenth century.

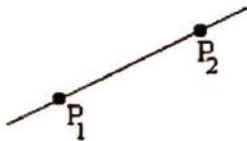
Modern creative origami came into existence thanks to Akira Yoshizawa. Beginning from the 1930s, he created many new designs, including animals and birds, equal in elegance to the old. His works were shown in the East and in the West. In recent years, many mathematicians and origami enthusiasts contributed to the development of origami. Their discoveries and contributions expanded the horizon for origami in art and in math.

To look at origami mathematically, it is best to find out its characteristics and what it can do. For geometrical operations using any tools, axioms are the most fundamental steps that can be done using those tools. The axioms for constructions using straightedge and compass, which is the one we use most frequently in high schools, are:

1. Given two points, we can make a line.
2. Given two nonparallel lines, we can locate their point of intersection.
3. Given a point and a length, we can draw a circle with radius of the length centered on the point.
4. Given a circle, we can locate the point(s) of intersection with another circle or line.

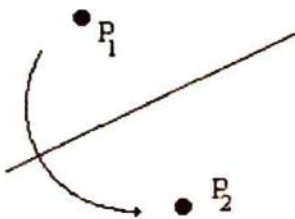
Now let's compare them with the six axioms of origami devised by Humiaki Huzita:

Axiom 1



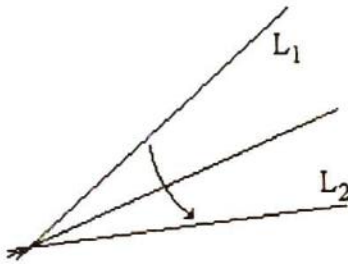
Given two points we can make a line (a crease or fold). (The same as SE&C axiom 1).

Axiom 2



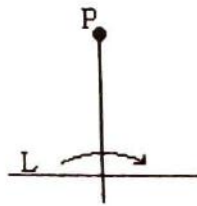
Given two points, we can make a fold that will map one point over the other. (This axiom finds the perpendicular bisector of the line segment formed by the two points)

Axiom 3



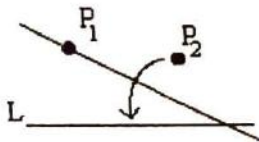
Given two lines, we can make a fold that will map one line over the other. (This axiom gives the angle bisector of the angle formed by the two lines).

Axiom 4

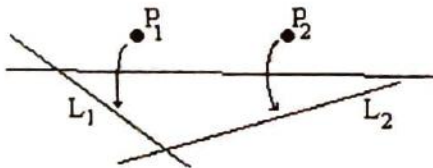


Given a point and a line, we can make a fold passing through the point that is perpendicular to the line.

Axiom 5



Given two points and a line, we can make a fold through one point that will map the other point onto the line. (This axiom locates the tangent through P_1 of the parabola with P_2 as the vertex and L as the directrix. Note: this axiom only works under the circumstance that the distance between the two points is not less than that between the first point and the line.)



Axiom 6

Given two points and two lines, we can make a fold that will map one point onto one line and the other point onto the other line. (This axiom finds the tangent of two parabolas. In general, the axiom is impossible to be reconstructed using straightedge and compass.)

Almost all origami axioms can be performed using one or more SE&C axioms, even axiom 5, which is equivalent to solving a second-degree equation. Axiom 6, however, is beyond the scope of any SE&C axioms, since it is equivalent to solving a third-degree equation. That is why this axiom is involved in many constructions, such as trisecting an angle and doubling the area of a cube, that would be impossible using SE&C.

Square paper has become widely used now for creative paperfolding because of its versatility. Other paper shapes however, still have their shares in origami. These include the different kinds of triangles, rectangles, regular pentagons, regular hexagons, regular octagons, and ovals. Because these shapes are rarely used, paperfolders usually make them from sheets of square paper. Some shapes are quite a challenge (some, like ovals, are impossible). One would be the Golden Rectangle, in which the ratio of the short side over the long side is phi ϕ , $(\sqrt{5} - 1)/2$. One of the best folds to produce the Golden Rectangle is shown on the right, in which midpoints R and C of the opposite

sides form a fold, the corner A and the midpoint C forms a fold, fold AS maps AB onto AC, and a perpendicular line through point S closes the rectangle. A simple geometric proof done by Higashi Abé shows its validity:

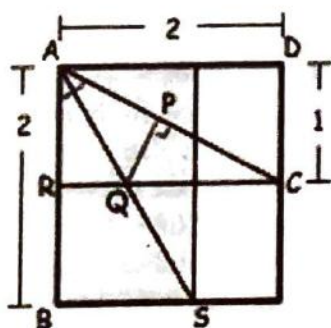


Figure 1 The Golden Rectangle

1. Assuming that the side of the square measures 2, according to the Pythagorean Theorem, $AC = \sqrt{5}$.
2. After dropping a perpendicular line QP from point Q to line AC, we can see that $\triangle ARQ \cong \triangle APQ$ according to the A.A.S. postulate. Consequently, $AP = 1$ and $PC =$

$$\sqrt{5} - 1.$$

3. $\triangle ADC \sim \triangle CPQ$ according to the A.A. postulate, $\therefore PQ = (\sqrt{5} - 1)/2 = QR$.
4. $\triangle ARQ \sim \triangle ABS$ according to the A.A. postulate, $\therefore SB = \sqrt{5} - 1$.
5. $SB : AB = (\sqrt{5} - 1)/2$, phi ϕ .

By manipulating angles and characteristics of the paper, it is possible to make triangles, pentagons, hexagons, octagons, and even decagons.

The only way to see the magic in each origami model is to experience it first hand. It's always important to keep your eyes open for the secrets hidden in every fold. Kazuo Haga, who was apt to find fascinating mathematical properties in simple folds, made a few discoveries in one fold that were later named the Haga Theorem by Kouji Fushimi. Using a square paper, folding a corner to the midpoint of the opposite side forms three triangles, which are all similar to each other and which have sides in the ratio of 3 : 4 : 5. Although not pointed out in the picture on the right, the exposed portion of the left edge of the square paper is actually twice as long as the part covered by the paper plus the dotted part. That means, the fold also divides the left side into lengths with ratio of 2 : 1. It is very easy to verify this theorem.

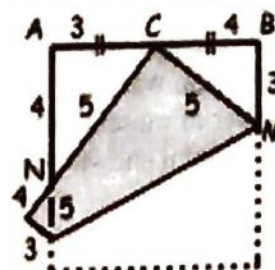


Figure 2 The fold demonstrating the Haga Theorem

1. Assume that the length of each side of the square paper is 1. Then $AC = CB = \frac{1}{2}$.
2. Let $BM = x$. CM would then be $1 - x$. By applying the Pythagorean Theorem, we get $x^2 + \frac{1}{4} = (1 - x)^2$. Now, we solve for x .

$$x^2 + \frac{1}{4} = 1 - 2x + x^2$$

$$2x = 1 - \frac{1}{4}$$

$$2x = \frac{3}{4}$$

$$x = \frac{3}{8}$$

3. Since $\angle NCM$ is a right angle, $\angle ACN \cong \angle CMB$ because they are both complementary angles of $\angle BCM$. $\therefore \triangle CAN \sim \triangle MBC$ according to the A. A. postulate.
4. Now we can figure out the length of AN using this proportion:

$$\frac{AN}{AC} = \frac{CB}{BM}$$

$$\frac{AN}{\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{3}{8}}$$

$$\frac{3}{8}AN = \frac{1}{4}$$

$$AN = \frac{2}{3}$$

This is just one of the methods used to divide the square paper into thirds. Using the properties of a square paper, we can also easily divide it into fifths and sevenths.

Even the simplest flat vertex fold can contain many surprises. Flat folds are defined as origami models that can be pressed without adding new creases. It also means that the faces in a flat fold are facing either of two opposite directions. They are the most basic origami models because they only contain mountain folds, shaped like Λ , and valley folds, shaped like V . A flat vertex fold is a model with one vertex. Below are two examples. The dashed lines and dotted-and-dashed lines represent valley folds and mountain folds respectively, and are used in standard origami books.

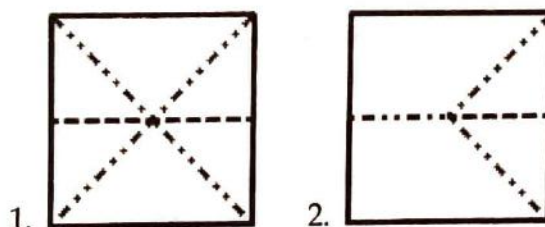


Figure 3 Flat Vertex Folds

After observing these flat vertex folds, some paperfolders were able to draw conclusions about their characteristics. One of them was Jun Maekawa, a Japanese physicist. His conclusion was later called Maekawa's Theorem. It states that the difference between the number of mountain folds and valley folds in a flat vertex fold is always 2. The proof of the theorem shown here was done by Jan Siwanowicz.

1. Let n = the number of folds around the vertex. Let M = the number of mountain folds and V = the number of valley folds. $n = M + V$.
2. After folding a flat vertex fold, the vertex becomes a point. We'll use the second one from the examples on the previous page. If we truncate the point and look at the fold from above, we see this polygon:



3. It is actually a flat polygon in which the angles measure either 0° or 360° . Angles formed by mountain folds are 0° , and angles formed by valley folds are 360° . n = the number of corners and the number of sides of the polygon.
4. Since the sum of interior angles = $(n - 2)180$, we obtain this equation:

$$0M + 360V = (M + V - 2)180$$

$$360V = 180M + 180V - 360$$

$$180M - 180V = 360$$

$$M - V = 2$$

More inferences can be derived from this theorem. A corollary states that the degree of the vertex in a flat vertex fold must always be even. Another one, discovered by Toshiyuki Meguro, states that the faces of the crease pattern of any flat fold are two-colorable, which means that no two faces with the same color will share a side. It is very

similar to the famous Four Color Theorem, which states that all maps can be colored with only four colors so that no regions with the same color will share borders.

Toshikazu Kawasaki discovered something else in a flat vertex fold. In what is called Kawasaki's Theorem, he states that the sum of odd-numbered angles and the sum of even-numbered angles around a flat vertex fold are both 180° . Using the second flat fold again, let's see this proof developed by Thomas Hull, which uses an interesting method.

1. Let A_i denote the angles around the vertex, where $i = 1, 2, 3, \dots, 2n$. Now let's look at the folded model through the eyes of an ant walking along its exterior starting from the fold furthest to the left.
2. As the ant walks on the paper, it will move along the angles, starting with A_1 . When it reached the second fold, it will switch direction and walk along A_2 (since this is a flat polygon, the ant only walks in two opposite directions. We'll say that toward the right is the positive direction, and toward the left is the negative direction). It will continue walking along all the angles until it comes back to the beginning. Below would be what its path looks like, stretched vertically a bit:



3. Since the ant is back to where it started, the resulting sum, or displacement, of all the angles is 0° . And because the ant walks in the positive direction on odd-numbered angles and in the negative direction on even-numbered angles, we obtain this equation:

$$A_1 - A_2 + A_3 - \dots + A_{2n-1} - A_{2n} = 0$$

4. The sum of these angles is 360° , however, because all of them are around a single vertex, so if we add the above equation to this one:

$$A_1 + A_2 + A_3 + \dots + A_{2n-1} + A_{2n} = 360$$

we get:

$$2A_1 + 2A_3 + 2A_5 + \dots + 2A_{2n-1} = 360$$

$$A_1 + A_3 + A_5 + \dots + A_{2n-1} = 180$$

5. If the sum of all odd-numbered angles is 180° , then the sum of all even-numbered angles must also be 180° .

By induction, we can say that Maekawa's Theorem and Kawasaki's Theorem apply to all vertices within the border of the paper in the crease pattern of any flat origami fold since they all must fold flat. However, it isn't always true that the result of all flat folds in a crease pattern is a flat origami fold. Just looking at a network of creases on a paper, it is very hard to imagine what shape will result from them. If we have a complete understanding of the collective behavior of all creases on a paper, we would be able to design our own models.

The study of origami helps us look at geometry from yet another point of view. The development of this field started fairly recently, so there isn't much material on it. Mathematicians and model designers have a long way to go to realize the full potential of origami.

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Some Fun Inequalities

By Joel Lewis

In most of high school mathematics, we deal solely with equalities: Find the value of x which makes y exactly equal to 3. If A equals $\langle 3, 4, 5 \rangle$ and B equals $\langle 1, 7, -2 \rangle$, then what does $A \times B$ equal? In this article, we will consider another type of relation, the inequality.

First, to the real numbers. For any real number t , it is always true that $t^2 \geq 0$. Equality holds, of course, only when $t = 0$. This is known as the *trivial inequality*. Despite its seeming obviousness (or for that matter, triviality), many not-so-obvious and wide-ranging facts can be proved using it and algebraic manipulation. Here are three examples:

1. Find all quadruples of real numbers, (a, b, c, d) , such that $10a^2 + 5b^2 + 37c^2 + 50d^2 = 6ab + 4bc + 60cd + 10da$ and $a + b + c + d = 90$.
2. Prove that if $x^2, y^2 \leq 1$ then $\sqrt{1-x^2} + \sqrt{1-y^2} \leq 2\sqrt{1-(\frac{x+y}{2})^2}$, and describe all pairs (x, y) for which equality holds.
3. Prove that the average of two non-negative numbers is greater than or equal to the square root of their product, and that equality holds if and only if the two numbers are equal.

This third problem represents one case of the *Arithmetic Mean - Geometric Mean Inequality*, usually shortened to AM-GM. The beauty of AM-GM lies in the fact that it can be generalized to any number of non-negative numbers: The arithmetic mean of n non-negative real numbers is greater than or equal to their geometric mean, with equality holding if and only if all of the numbers are equal. (That is, their sum divided by n is greater than or equal to the n th root of their product.)

Proof: We will prove the statement using a fairly complicated induction argument. Our base case is $n = 2$, and we have already shown that it works. We will now use it to prove that for n equal to any power of 2, AM-GM holds:

Assume that the AM-GM inequality is true for $n = 2^k$. In other words, for all sets of non-negative reals, $\{a_1, a_2, \dots, a_{2^k}\}$ we have $\frac{a_1 + a_2 + \dots + a_{2^k}}{2^k} \geq \sqrt[2^k]{a_1 a_2 \dots a_{2^k}}$, with equality holding only when all the a_i are equal.

Now, we can take a set of non-negative reals $\{b_1, b_2, \dots, b_{2^{k+1}}\}$ such that the average of $b_{2^{k-1}+1}$ and b_{2^k} is a_i for each i between 1 and 2^k . The a_i can be any set of non-negative real numbers, so the b_i can as well (since any set of b_i has a corresponding set of a_i). Now, let us plug our b_i into the AM-GM we have assumed for the a_i :

$$\frac{\frac{b_1 + b_2}{2} + \frac{b_3 + b_4}{2} + \dots + \frac{b_{2^{k-1}+1} + b_{2^k}}{2}}{2^k} \geq \sqrt[2^k]{\left(\frac{b_1 + b_2}{2}\right) \left(\frac{b_3 + b_4}{2}\right) \dots \left(\frac{b_{2^{k-1}+1} + b_{2^k}}{2}\right)}$$

Applying the $n = 2$ case of AM-GM to each term on the right side of this inequality and simplifying, we obtain:

$$\frac{b_1 + b_2 + \dots + b_{2^{k+1}}}{2^{k+1}} \geq \sqrt[2^k]{b_1 b_2} \cdot \sqrt[2^k]{b_3 b_4} \dots \sqrt[2^k]{b_{2^{k-1}+1} b_{2^k}} = \sqrt[2^{k+1}]{b_1 b_2 \dots b_{2^{k+1}}}$$

So by induction, we know that AM-GM is true for any set of 2^k non-negative integers for some k .

Now, if we show that the truth of AM-GM for n implies it for $n-1$, we will have proven the inequality entirely, because any number k can be reached by picking a large power of 2 and then subtracting 1 repeatedly, so our induction argument will reach all positive integers.

So, let us assume that for all sets of n non-negative numbers,

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}.$$

We can let a_n be the average of the other $n-1$ numbers, $\frac{a_1 + a_2 + \dots + a_{n-1}}{n-1}$. Then

$$\frac{a_1 + a_2 + \dots + a_{n-1} + \frac{a_1 + a_2 + \dots + a_{n-1}}{n-1}}{n} \geq \sqrt[n]{a_1 a_2 \dots a_{n-1} \left(\frac{a_1 + a_2 + \dots + a_{n-1}}{n-1} \right)}.$$

We can simplify the left side and raise both sides to the n^{th} power to get:

$$\left(\frac{a_1 + a_2 + \dots + a_{n-1}}{n-1} \right)^n \geq a_1 a_2 \dots a_{n-1} \cdot \frac{a_1 + a_2 + \dots + a_{n-1}}{n-1}.$$

We can divide by the shared factor on both sides and then take the $(n-1)^{\text{th}}$ root to get,

$$\frac{a_1 + a_2 + \dots + a_{n-1}}{n-1} \geq \sqrt[n-1]{a_1 a_2 \dots a_{n-1}},$$

which is exactly what we wanted! Therefore, the AM-GM holds generally.

We now give a few problems using the AM-GM inequality:

4. Go back through the proof of the general case and verify that equality holds only when all the a_i are equal.

5. Prove $(x + y)^n \geq 2^n (xy)^{\frac{n}{2}}$.

6. Use the AM-GM inequality to prove the AM-GM-HM inequality, where HM is the harmonic mean:

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n} \geq \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}},$$

if each a_i is positive, and equality holds if and only if all of the a_i are equal.

7. Find the maximum value of A , in terms of n , such that the following inequality holds for all integers n and all positive real numbers x : $x^n + x^{n-2} + \dots + x^{2-n} + x^{-n} \geq A$, and find the values of x such that we have equality.

8. Find the least possible value of $(a_1 + a_2 + a_3 + a_4) \cdot (\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \frac{1}{a_4})$, where each a_i is positive. Then, generalize to $(a_1 + a_2 + \dots + a_n) \cdot (\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n})$.

The third inequality we will consider is the *Cauchy-Schwarz Inequality*, sometimes known as the *Cauchy-Buniakowski Inequality*. It is associated with the verb "to Cauchy", as in, "Hey man, that problem was so easy - you just have to Cauchy the left side and then it follows trivially!"

The Cauchy-Schwarz Inequality states that for $2n$ real numbers, we always have $(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \geq (a_1b_1 + a_2b_2 + \dots + a_nb_n)^2$, with equality holding only when $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \dots = \frac{a_n}{b_n}$. The proof given here is just one of many possible, and the reader is invited to try and prove it independently.

Proof: Let $A = a_1^2 + a_2^2 + \dots + a_n^2$, $B = a_1b_1 + a_2b_2 + \dots + a_nb_n$, $C = b_1^2 + b_2^2 + \dots + b_n^2$, and assume not all of the a_i are zero (since in that case the inequality is obviously true) so that $A > 0$. Consider $f(x) = Ax^2 + 2Bx + C$.

We can rewrite $f(x)$ by separating the coefficients of x and then collecting terms by subscripts:

$$(*) \quad f(x) = (a_1^2x^2 + 2a_1b_1x + b_1^2) + (a_2^2x^2 + 2a_2b_2x + b_2^2) + \dots + (a_n^2x^2 + 2a_nb_nx + b_n^2)$$

Each term of this sum is a perfect square, so by the trivial inequality, $f(x)$ is greater than zero for any choice of x . In particular, we may choose $x = -\frac{B}{A}$.

$f(-\frac{B}{A}) = \frac{B^2}{A} - 2\frac{B^2}{A} + C = \frac{AC - B^2}{A} \geq 0$. Since $A > 0$, $AC - B^2 \geq 0$ and $AC \geq B^2$, which is exactly Cauchy-Schwarz. Furthermore, equality holds only when each of the terms in equation $(*)$ are equal to 0. From this, we see that equality holds only when $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \dots = \frac{a_n}{b_n} = -x$.

9 Use Cauchy-Schwarz to prove the result of question 8.

10. Use Cauchy-Schwarz to prove that the quadratic mean of n numbers, $\sqrt{\frac{a_1^2 + a_2^2 + \dots + a_n^2}{n}}$, is greater than or equal to the arithmetic mean, with equality holding only when all the a_i are equal.

11. Prove that, for any real number q , we have

$$(a_1^qb_1^q + a_2^qb_2^q + \dots + a_nb_n^q) \cdot (a_1^{2-q}b_1^{2-q} + a_2^{2-q}b_2^{2-q} + \dots + a_n^{2-q}b_n^{2-q}) \geq (a_1b_1 + a_2b_2 + \dots + a_nb_n)^2$$

and

$$(a_1^qb_1^{2-q} + a_2^qb_2^{2-q} + \dots + a_nb_n^{2-q}) \cdot (a_1^{2-q}b_1^q + a_2^{2-q}b_2^q + \dots + a_n^{2-q}b_n^q) \geq (a_1b_1 + a_2b_2 + \dots + a_nb_n)^2$$

so long as those expressions are defined.

Finally, we will consider one last inequality, but this is for the complex numbers:

Given any two complex numbers, a and b , $|a + b| \leq |a| + |b|$, where $|x + iy| = \sqrt{x^2 + y^2}$, x and y real, or in other words the distance of the point in the complex plane from the origin.

For real numbers this inequality is almost trivially true. If a and b have the same sign, then equality holds. If they have different signs, the left side will become a positive number plus a negative number and the right side will become the same positive number plus a positive number, which is clearly larger. For the case of complex numbers, the situation is much more complicated. The simplest proof is purely geometric in form. Consider the complex plane and two complex numbers, a and b . We know that the sum, $a + b$ can be located in the plane by forming the parallelogram three of whose vertices are 0, a , and b . $a + b$ will be the vertex opposite 0, as shown. One edge of the parallelogram has length $|a|$, and the other has length $|b|$. The diagonal has length $|a + b|$. We know from geometry that the sum of the lengths of two sides of a triangle is larger than the third length; therefore we have $|a + b| \leq |a| + |b|$, just like we wanted. For this reason, the inequality is known as the *triangle inequality*.

Imaginary Axis

$a + b$
 b
 a
Real Axis

Examples:

12. Prove that $\sum_{i=1}^n |z_i| \geq \left| \sum_{i=1}^n z_i \right|$, or in other words that the sum of the absolute values of any n

complex numbers is greater than or equal to the absolute value of their sum.

13. Describe the conditions under which equality holds in problem 12.

14. Prove $|a - b| \leq |a| + |b|$ for two complex numbers a and b .

Many of the questions in this article and the proof of the Cauchy-Schwarz Inequality come from The USSR Olympiad Problem Book, by D. O. Shklarsky, N. N. Chentzov, and I. M. Yaglom, translated by John Maykovich. It was published in America by Dover Publications. It contains many problems with solutions from geometry, algebra and number theory and is an excellent resource for anyone who has an interest in problem solving.

Monte Carlo Simulation for Stochastic Processes: Call Option Valuation

By Revaz Surguladze

Introduction

Random events, such as those described in this article, occur in the real world. In fact, the complexity of the real world is due entirely to the fact that it is the result of a series of completely random events. Very few (if any) of these real world phenomena can be described via exactly solvable mathematical models. In a case when it is not possible, mathematical simulation methods, such as Monte Carlo simulation, are used to describe and quantify these processes.

Few basic definitions are in order. An unknown quantity, x that can take on a finite set of ordered values, $x_1, x_2, \dots, x_n$, where the probability of occurrence of x_i , is p_i , is known as a *random variable*. Often, the set of possible values for a random variable is displayed graphically, as a *probability distribution*, or *density*. In a probability distribution, the abscissa indicates the possible values of the random variable, while the ordinate shows the probabilities, or frequencies of occurrence for each value. The *mean* and *standard deviation* of a set of values are key determinants of the shape of the distribution of the set. The mean, or *expected value*, of a set of values for a random variable, x , is denoted by $E(x)$, where operation $E(x)$ is defined as

$$E(x) = \sum_{i=1}^n x_i p_i \quad (1)$$

where p_i is the probability that x_i will occur. Thus, the expected value of a set of possible outcomes of a random variable can be defined as the probability weighted average of all possible outcomes for that variable. The standard deviation of a set of values is a measure of the dispersion, or deviation, of these values from the mean value of the set. Standard deviation for random variable x , denoted by σ_x , is defined as follows.

$$\sigma_x = \sqrt{E[(x - \bar{x})^2]}. \quad (2)$$

where $\bar{x} = E(x)$. This can be further simplified as follows

$$\sigma_x = \sqrt{E(x^2) - 2E(x)E(x) + E(x)^2} = \sqrt{E(x^2) - E(x)^2}. \quad (3)$$

These two quantities, $E(x)$ and σ_x , are the main characteristics of their corresponding probability distribution. One particular probability distribution is the *standardized normal distribution* (Fig. 1). The mean for this distribution is 0, while the standard deviation is 1.

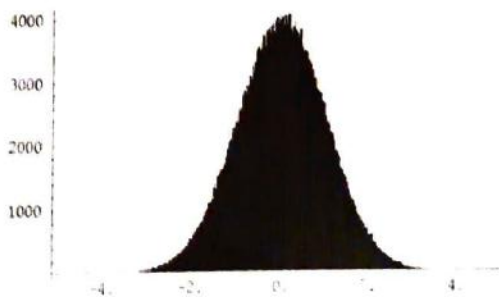


Fig. 1. Normal distribution with mean 0 and standard deviation 1. (generated with 1 million random points using MATHEMATICA program)

In other words, random picks from a set of normally distributed numbers will have an average of zero. Furthermore, it can be shown that there is about a 67% chance that the randomly picked number will be within the $-1,1$ interval, about a 95% chance that it will be within the $-2,2$ interval, about 99% that it will be within the $-3,3$ interval, and so on.

A model that is typically used to simulate the behavior of stock prices over a period of time is the *geometric Brownian motion*. This model states that the relative change in values over time can be described by the following relationship

$$\frac{\Delta S_t}{S_{t-1}} = \mu \Delta t + \sigma \varepsilon \sqrt{\Delta t} \quad (4)$$

where S_{t-1} is the value of a random variable (stock price, in our case) at time $t-1$, and ε is a *standardized normally distributed random variable*; that is, it has a mean zero and a standard deviation of one. To understand meanings of μ and σ , set $\Delta t = 1$, say, 1 year. Then, taking the expectation $E()$ in the both sides of eq.(4), one finds that $E(\Delta S_t / S_{t-1}) = E(\mu + \sigma \varepsilon) = E(\mu) + \sigma E(\varepsilon) = \mu$, since $E(\varepsilon) = 0$. That is, μ is an annualized relative change in stock price. Similarly, starting from the definition (2),

$$\begin{aligned} \sigma^2_{\Delta S/S} &= E[(\mu + \sigma \varepsilon - E(\mu + \sigma \varepsilon))^2] = E[(\mu + \sigma \varepsilon - \mu)^2] = E[(\sigma \varepsilon)^2] = E[\sigma^2 \varepsilon^2] = \sigma^2 E[\varepsilon^2] \\ &= \sigma^2 E[(\varepsilon - E(\varepsilon))^2] = \sigma^2 \end{aligned}$$

since $E[(\varepsilon - E(\varepsilon))^2] = 1$, because the standard deviation for ε is 1. One concludes that the random variable $\Delta S/S$, the relative change in stock price in one year, is normally distributed with a mean μ and standard deviation of σ . Δt could have been any other time interval.

According to the above model, the relative change in the value of stock price $\Delta S/S$ (a random variable) is the sum of the deterministic linear function of time $\mu \Delta t$ (sometimes called drift) and the random component $\sigma \varepsilon \sqrt{\Delta t}$. The model indicates that the drift is a linear function dependent on the expected value and the length of the time interval involved. However, there may occur certain random deviations from this linear change – drift. The standard deviation of these random changes is $\sigma \sqrt{\Delta t}$.

The above process in eq.(4) may be studied via Monte Carlo simulation, where a computer program, in this case, MATHEMATICA, generates standardized normally distributed random numbers ε , which are then used in eq.(4) to generate the particular path of the relative change in the stock price for a certain time horizon. Fig.2 shows one possible path of the stock price.

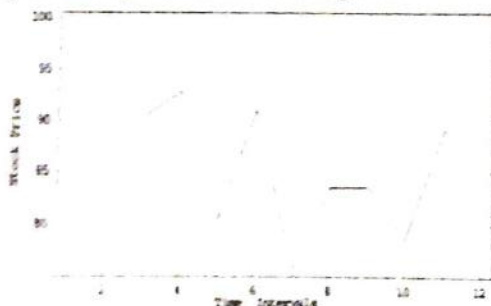


Fig. 2. A sample path of a stock price with an initial price of 100 within the 12-month horizon

In fig.3, Monte Carlo simulations of multiple scenarios for the stock price paths have been plotted.

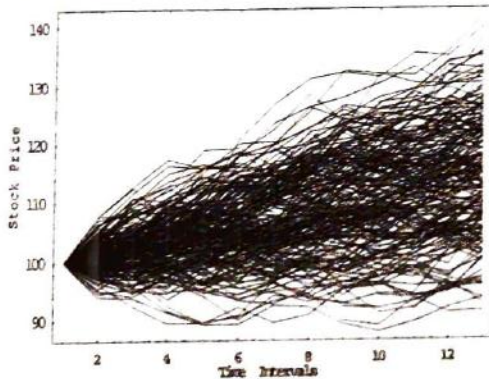


Fig. 3 Monte Carlo price path simulations for stock price, with initial value of \$100, standard deviation 30%, and 12 time intervals

Applying Monte Carlo

Monte Carlo simulation methodology is widely used in solving various problems where analytical solutions cannot be found without reliance on undesirable approximations. In the present work, Monte Carlo simulation methodology is used to evaluate a *call option* contract, which is widely traded in the world's securities markets. (A call option gives the buyer the right, but not obligation, to buy a share of an underlying stock in the future time for a predetermined price.)

Consider an example. A company's stock currently sells for $S_0 = \$100$ a share, and has a volatility of $\sigma = 30\%$ per year (recall σ in eq.(4)). A call option on this stock would give the right to buy this stock at a later time, say, in $T = 1$ year (call option expiration) for, say, $X = \$100$. Would such a contract be of any value to its buyer and what would be a fair price?

Intuitively, a call option contract would have a nonzero value, because the stock price in one year is not a deterministic quantity. Indeed, there is always a chance that the price of the company's stock in one year will end up above \$100 – an obvious profit potential for the contract holder. Namely this “*chance of profit*” gives the nonzero value to the call option. However, how much should one be willing to pay for this “*chance of profit*” today?

In other words, one would like to quantify the uncertainty in the future stock price in dollar terms – that is value the risk in the underlying stock price. In contrary, if one knows for certain that the company's stock in one year will be at \$100, then, obviously, the right, given by the call option, to buy the stock for \$100 in one year will have no value.

Assume that the stock price at the end of one year (expiration of the call option) is S_T . Then, according to the contract definition, the payoff to the holder at the expiration will be

$$\text{Max}[S_T - X, 0].$$

However, because S_T is a probabilistic quantity, the above payoff is a probabilistic quantity as well. Therefore, one may calculate the expected value of the call option payoff at the expiration as follows.

$$E[\text{Max}(S_N - X, 0)]$$

This is how much the contract holder should expect to be paid off in one year *on average*, in the long run (after many identical call option contracts). However, today's value of the amount received in one year is obviously less than the amount itself. Indeed, for example, everyone would prefer to be paid \$100, or even \$99.95, today than in one year. In practice, in order to find today's value of a future cash flow, one discounts the future cash flow with the present *risk free rate of interest* (determined from the yield on investments in US Government securities). Therefore, the present value of the call option expected payoff, which in turn is the call option price, is computed as follows.

$$C = e^{-rT} E[\text{Max}(S_T - X, 0)] \quad (5)$$

where r is the risk free rate of interest. Eq.(5) is the main working equation.

Next, using the Mathematica program, one simulates the stock price S_T at the end of one year period ($T=1$) using eq.(4) to compute the increments in stock price for monthly intervals (see fig. 3).

$$S_t = S_{t-1} + \Delta S_t = S_{t-1} + rS_{t-1}\Delta t + \sigma S_{t-1}\sqrt{\Delta t} = S_{t-1}(1 + r\Delta t + \sigma\epsilon\sqrt{\Delta t}) \quad (6)$$

(It suffices to set $\mu = r$, because the option price is, in fact, independent of μ).¹ Having all S_T s computed, one evaluates $\text{Max}(S_T - X, 0)$ for every S_T and then averages them (computes the expected value $E[\]$). Finally, one finds the present value by multiplying the computed expected future value by e^{-rT} .

One also computes the standard deviation of the call price itself. Indeed, since one is using the finite number of Monte Carlo simulations, the obtained answer for the call price deviates from the true answer. The "risk" of deviation, in other words, the computational risk is described by the standard deviation computed as follows.

$$SD_C = \frac{1}{n-1} \sqrt{n[E(C^2) - E(C)^2]} \quad (7)$$

where n is the number of S_T simulated. This indicates that standard deviation (risk of a deviation from the correct answer) decreases as the number of simulations, n , increases.

One also computes the standard error of the mean as follows

$$SE = \frac{SD_C}{\sqrt{n}} \quad (8)$$

This characterizes how far the computed mean (result) is from the true mean of the distribution.

For the actual computation, the following values of parameters have been used: the initial stock price $S_0=100$, strike price $X=100$, volatility $\sigma=0.3$ (30%), time to expiration $T=1$, and risk free rate of interest $r=0.05$ (5%).

¹ This can be shown by just computing the call price for different values of μ .

With a function created in the Mathematica system to compute the call option price, and using the above input values and $n=1$ million Monte Carlo points, one obtains the following result for the call price

$$C = 14.200 .$$

The corresponding standard deviation is

$$SD = 0.021$$

and the standard error of the mean is

$$SE = 0.000021 .$$

Recalling the definition of the standard deviation, one concludes that there is about a 67% chance that the true answer is within the interval

$$14.20-0.021, 14.20+0.021.$$

There is about 95% chance that the true answer is within the interval

$$14.20-2 \times 0.021, 14.20+2 \times 0.021.$$

There is about 99% chance that the true answer is within the interval

$$14.20-3 \times 0.021, 14.20+3 \times 0.021.$$

In fig 4, one plots the computation results for different numbers of Monte Carlo simulations.

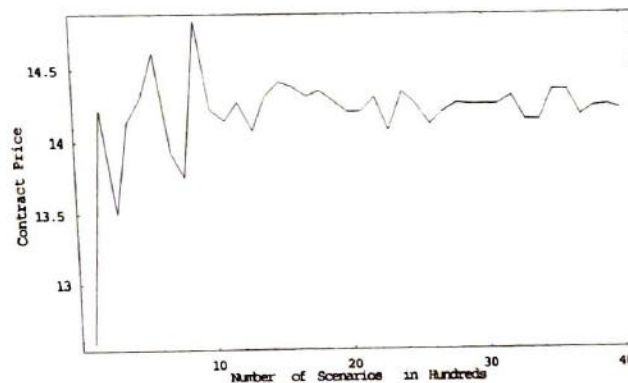


Fig. 4 Call price computed with successively higher numbers of Monte Carlo simulations.

One can see that as the number of simulations increases, the oscillation around horizontal line stabilizes computation result for the call price gradually converges.

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Fermat and his Contributions to Mathematics

By Yuetian (Peak) Xu

Pierre de Fermat was a great mathematician of his time, who was called "Prince of Amateurs" by E. T. Bell. He was not even included by Julian Coolidge in his *Mathematics of Great Amateurs* because Coolidge believed that his immense talent qualified him as a professional. Fermat made many important contributions to mathematics in field of probability, calculus, and number theory including his famous Last Theorem.

He was born on August 20, 1602 in Beaumont-de-Lomagne, a small town in southwestern France. He was educated at a Franciscan monastery of Grandseve and then briefly at the University of Toulouse. He later served as a councilor in the judiciary and other governmental posts. As an official, he was discouraged from extensive social contacts as to avert favoritism. Therefore, he spent most of his leisure time in pursuing mathematics.

Fermat along with Blaise Pascal co-founded the mathematical branch known as probability theory. This was started when a Parisian gambler, Antoine Gombaud, Chevalier de Méré asked Pascal a question concerning a game called Points in which players win points according to rolls of a dice and the first player to win a set amount of points is the winner. Gombaud and another gambler were participating in a game when they had to abandon it due to pressing matters. He asked Pascal concerning the mathematics of the game as to divide the prize money fairly assuming the game continued with equal opportunity for players to gain points. Pascal then corresponded with Fermat upon the matter and then jointly solved the problem as well as formulating a set of laws of probability as to calculate mathematically chances of various events occurring. This can result in some counterintuitive results. For example, what would the probability of two people in a class of 23 people sharing the same birthday? Most people might guess a probability of 10% or so by intuition. In reality, the probability is approximately 50.7%. While 23 out of 365 seem small, one must remember that we are not considering the number of individuals but rather the number of pairing of individuals.

Therefore, there are $\frac{23 \cdot 22}{2} = 253$ pairs of individuals. Then, from that point, the probability is no longer so surprising. Thus, the mathematics of probability is interesting as well in being counterintuitive in many circumstances. Fermat, progenitor of this branch of mathematics, made everything from stock speculation to calculation of insurance rates possible.

Another area of math that Fermat helped in discovering is the field of calculus. While Newton and Leibniz were widely credited with discovering calculus, Fermat's work was certainly influential to say the least. In 1934, Louis Trenchard Moore discovered a note by Isaac Newton that stated that he based his work upon "Monsieur Fermat's record of drawing tangents". Indeed, this showed how Fermat helped create a branch of math that allowed us to prove numerous physical formulas and calculate the rates of change of various things, as the world we live in is quite dynamic.

But perhaps his most important contributions in math are in the area of number theory. Naturally, his Last Theorem was judged by many to be the most celebrated considering the fact that it baffled many for more than 350 years. When it was proved by Andrew Wiles, the proof introduced many new techniques and concepts that is highly useful for the future. Another important theorem is Fermat's Little Theorem which states that $a^{p-1} \equiv 1 \pmod{p}$ for all numbers a not congruent to 0 mod p and all prime numbers p . This theorem is also extremely useful in many cases dealing with number theory problems. In fact, Fermat left behind such a multitude of theorems and notes in his copy of *Arithmetica* that his son Clément-Samuel published his annotations in a special edition of *Arithmetica*. Even before his death, he used to tease other mathematicians by challenging them to prove his theorems. One such mathematician Wallis had to give up out of frustration and called Fermat "that damned Frenchman".

Perhaps even more interesting than his theorems are some of the techniques that he pioneered. One particularly powerful technique is the method of infinite descent. An example below illustrates the proof of irrationality of $\sqrt{2}$ by that method.

Proof:

First, assume that $\sqrt{2}$ is rational, thus it can be written as the quotient $\frac{a_1}{b_1}$ where a_1 and b_1 are positive integers.

Then, by use of the difference of squares, we can use the following equality,

$$\frac{1}{\sqrt{2}-1} = \sqrt{2} + 1$$

Substituting $\sqrt{2} = \frac{a_1}{b_1}$ into the left side and then solving for $\sqrt{2}$ on the right side yields the following,

$$\sqrt{2} = \frac{2 - \frac{a_1}{b_1}}{\frac{a_1}{b_1} - 1}$$

Multiplying top and bottom by b_1 ,

$$\sqrt{2} = \frac{2b_1 - a_1}{a_1 - b_1}$$

Since $1 < \frac{a_1}{b_1} < 2$, both the numerator and denominator are positive integers. Thus, the numerator and denominator can now be represented by positive integers a_2 and b_2 . Obviously, $a_1 > a_2$ and $b_1 > b_2$. Thus, the same process can be repeated an infinite number of times and thus there is an infinite descent of integers $a_{n+1} < a_n$ and $b_{n+1} < b_n$ each time. Therefore, there is no smallest positive integers a and b such that their quotient is $\sqrt{2}$, creating a contradiction and thus $\sqrt{2}$ is irrational.

Fermat used this method with great efficiency, especially when he proved that $x^4 + y^4 = z^2$ has no solutions among integers and consequently the $n=4$ case of his last theorem. The proof goes as follows.

Proof:

Assume integer solutions do exist, then the equation can be rewritten as following,

$$(x^2)^2 + (y^2)^2 = z^2$$

Therefore, x^2 , y^2 , and z is a Pythagorean triple and can be rewritten as the following,

$$x^2 = p^2 - q^2$$

$$y^2 = 2pq$$

$$z = p^2 + q^2$$

$2pq$ is a square, so either p or q is even and p^2 can be rewritten as a Pythagorean triple.

$$x^2 + q^2 = p^2$$

Rewriting x , p , and q gives the following,

$$x = r^2 - s^2$$

$$p = r^2 + s^2$$

$$q = 2rs$$

Since $2pq$ is a square, we can let $q = 2u^2$ and $p = v^2$, yielding the following,

$$2u^2 = 2rs$$

Now let $r = g^2$ and $s = h^2$, combine this with the fact that $p = v^2$, we obtain:

$$v^2 = g^4 + h^4$$

Which is the same form as our original equation with $v < z$. This process can be repeated ad infinitum, again producing an infinite descent through all positive integers, with no smallest solutions, contradicting the original assumption, proving that there are no solutions. Indeed, this method of infinite descent is quite a powerful technique that Fermat frequently employed.

However, one must use this with caution as Fermat had a little too much faith in his method of infinite descent when he stated that $2^{2^n} + 1 = p$ where p is a prime. This

was disproved by Euler for the $n=5$ case as $2^{2^5} + 1 = 4294967297 = 641 \times 6700417$.

In conclusion, Fermat made many wonderful contributions to mathematics in many fields with numerous theorems and techniques. His rich legacy certainly makes him deserving of the title "Prince of Amateurs."

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Hyperbolic Geometry

By Svetlana Goldenberg

Euclidian geometry was based on the five postulates stated Euclid's "Elements".

1. There is only one line that passes through any two given points on the plane
2. A line segment can be extended infinitely
3. Any point on the plane can be a center of a circle with any given radius
4. All right angles are congruent
5. There is only one line parallel to the given line and passing through the point that is not on a given line.

The fifth postulate was not accepted as an axiom for several centuries. Many mathematicians, including Gauss and Bolyai, attempted to prove it and eventually concluded that there was a type of geometry different from Euclidian. In fact, there are several types of non-Euclidian geometry such as geometry of the sphere, geometry of perspective, geometry of space, and other geometries that involve three dimensions. These geometries were sufficient for our three-dimensional world, but not for the outside space. Almost at the same time Lobachevsky, Bolyai, and Gauss invented the hyperbolic geometry, sometimes called four-dimensional geometry. The hyperbolic geometry is based on the space curved in a shape of a pseudosphere (sometimes called the space-time) and first four Euclidian postulates. The hyperbolic geometry is used in theories that explain the field created by the black holes and would be useful for navigation in space.

Three mathematicians are accredited for the discovery of hyperbolic geometry: Gauss, Bolyai, and Lobachevsky. They worked separately, but came up with similar results almost at the same time.

Gauss had been thinking about non-Euclidian geometries since the age of fifteen. He worked in the same area as Janos Bolyai but did not have enough confidence to publish his work. In 1823 Bolyai finished the treatise on hyperbolic geometry. He was in correspondence with Gauss and was shocked when he learned that Gauss, who was one of his friends, arrived to the same results earlier. Eventually, Janos Bolyai published his essay in 1832 as an appendix to his father's treatise. Nine years before that, in 1823, Lobachevsky completed his work on hyperbolic geometry. Unlike Gauss and Bolyai, Lobachevsky did not attempt to prove the fifth postulate as a theorem. Instead, he considered the cases when the fifth postulate does not necessarily hold. He characterized Euclidian geometry as a special case of a more general geometry. In his work *Geometria*, Lobachevsky outlined the basic concepts and principles of hyperbolic geometry.

Another mathematician who contributed to the development of hyperbolic geometry was Poincaré. He suggested a method of representation of pseudosphere that made it easier to understand and offered a more convenient way to prove theorems.

CURVED SPACETIME



Hyperbolic geometry can be applied to the hyperbolic space that has a shape of a pseudo sphere. The pseudo sphere is a curved space, but when discussing two-dimensional objects, it can be viewed as a curved plane with hyperbolic axes. The

pseudospherical plane would have a shape of a hyperbolic paraboloid. In the curved plane the shape of two-dimensional objects is distorted. Although the hyperbolic plane is different from the Euclidian plane, the two geometries have something in common. In hyperbolic geometry the fifth postulate is not held, but the other four postulates are true. Thus all properties and theorems that can be proved in Euclidian geometry without using the fifth postulate can be used in hyperbolic geometry.

In hyperbolic geometry as well as in Euclidian geometry the line segment is the shortest distance between two points, though in the curved plane the segment is curved. Two points determine a unique line and therefore there is only one line that passes through any two given points. However, the following statements that are true in Euclidian geometry are false or not necessarily true in hyperbolic geometry.

The Hyperbolic Paraboloid $(\frac{1}{9})x^2 - (\frac{1}{9})y^2 = z^2$



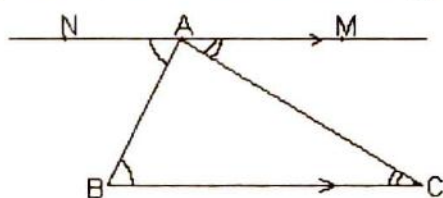
In Euclidean Geometry, lines are infinite and do not have a boundary (a point that they are headed toward, yet never reach). In the pseudosphere, however, the hyperbolic lines approach the boundary circle but never reach it. This provides the basis for the Poincaré Model. The Poincaré Model (also sometimes called the disk model) of the hyperbolic geometry represents a section of the pseudosphere. As a point becomes closer to the boundary circle, its distance from the center approaches infinity. The three circles below are the Poincaré Models. The first one shows a set of straight line segments in hyperbolic geometry of the same length. All the segments share a common endpoint, which is not located in the center of the boundary circle. The closer the endpoint of the segment is to the boundary circle, the shorter the segment appears. It might seem that the endpoints of some segments lie on the boundary circle, but they can only approach it.

The second Poincaré Model shows a set of straight hyperbolic line segments that have a common endpoint A located in the center of the circle. Length of segment AB is half of the length of segment AC. The length of AC is half of the length of the segment AD. Going counterclockwise each segment is twice as long as the previous segment. It appears that segments AG and AJ have the same length. However, length of AG is half of

the length of AJ. In this Poincaré Model as the lengths of the line segments increase the endpoints of the line segments approach the boundary circle. Also as the endpoints of the segments approach the boundary circle, the difference between the lengths of the segments appears to decrease.



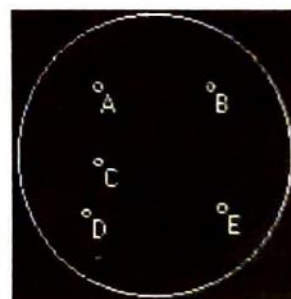
The third model is an example of a tessellation of congruent triangles in the pseudosphere. It shows that in hyperbolic geometry the shape of the triangle is distorted.



In Euclidian geometry the fifth postulate is used to prove that the sum of the angle measures of a triangle is equal to 180° . A line parallel to BC is drawn through point A to show that the sum of the angle measures of the

triangle is equal to the measure of the straight angle and therefore equal to 180° . In Euclidian geometry it is possible, because according to the fifth postulate, there is only one line that passes through point A and is parallel to line BC. In hyperbolic geometry, however, there are an infinite number of lines parallel to a given line and passing through a given point.

For example, in the Poincaré Model, hyperbolic lines AB and CB are parallel to hyperbolic line DE, since neither line AB nor line CB intersects line DE. Both hyperbolic lines AB and CB pass through point B. In Euclidean Geometry, if two lines are parallel to a third line, then the two lines are parallel to each other. In hyperbolic geometry two hyperbolic lines that are parallel to the third hyperbolic line might intersect.



Thus, in hyperbolic geometry, there are infinite number of lines that are parallel to line BC and pass through point A. However, In Hyperbolic Geometry, however, there does not exist any line such that both: $m \angle NAB \neq m \angle ABC$ and $m \angle MAC \neq m \angle ACB$.

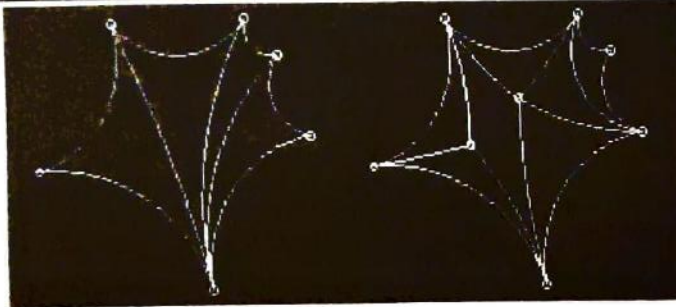
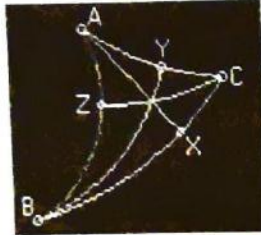
In hyperbolic geometry the angles are formed by the hyperbolic rays. The measure of such angle is equal to the measure of a Euclidian angle formed by the lines tangent to the sides of the angle at the vertex of the angle.



Thus, the sum of the measures of the angles of a triangle is always less than 180° . The difference between 180° and the sum of the measures of the

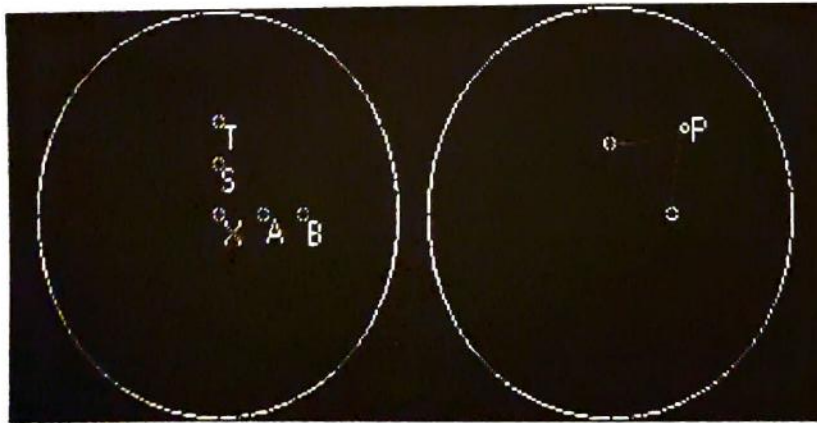
angles of a triangle is the defect of the triangle that could be used to determine its area. As the lengths of the three sides of a triangle decrease, the defect approaches 0.

In Euclidian geometry the area of the triangle is equal to half of the product of the lengths of the base and the altitude of the triangle. However, in hyperbolic geometry, this formula cannot be used. In the hyperbolic triangle ABC, length of base AB is 3.3, altitude CZ is 1.1, and $m\angle CZA = m\angle CZB = 90^\circ$. Length of base AC = 2.1, altitude BY = 2.5, and $m\angle BYA = m\angle BYC = 90^\circ$. The product of lengths of segments AB and CZ is not equal to the product of



lengths of segments AC and BY. The formula for the area of the triangle should always lead to the same result for the same triangle. The formula used in hyperbolic geometry is a linear function of the defect of the triangle. This formula could be used to find area of the polygon.

Any polygon can be divided into a finite number of non-overlapping triangles. The area of the polygon depends on the defect of the polygon, which is equal to the sum of defects of the triangles it was divided into. As the number of the triangles increases, the lengths of their sides decrease. Thus, as the number of the triangles increases, the defect of each of the triangles decreases. The sum of the defects would remain constant for the same or congruent polygons. The area of the polygon would be then equal to the sum of the areas of the triangles. The area and the defect of a polygon do not change if the polygon undergoes a translation.



In hyperbolic geometry translations are possible as well as in Euclidian geometry. Like in Euclidian geometry, in hyperbolic geometry there is an XY-coordinate plane. Like the Cartesian plane, it has two

axes that are perpendicular to each other and divide the plane into four quadrants. On both Cartesian and hyperbolic coordinate planes every point is defined by a unique set of coordinates that indicate the distance of the point from each of the axes. In both systems the distance from the point to the line is the perpendicular drawn from the point to the line. In the Cartesian plane, when the perpendiculars are drawn from the point to the axes, they form a rectangle with the axes. In hyperbolic geometry, rectangles do not exist.

Since the sum of the measures of the angles of the triangle is less than 180° , the sum of the angles of a quadrilateral is always less than 360° . Thus, if a quadrilateral has three right angles, the fourth angle must be acute. The model shows the coordinate plane in hyperbolic geometry, where point X is the origin. The lengths of segments XA and XS are 0.5 units. The lengths of segments XT and XB are 1.0 units. The perpendicular to the Y-axis at point T and the perpendicular to the X-axis at point B do not intersect. Then, according to the principle of the Cartesian plane, point with coordinates (1,1) would be undefined. In hyperbolic geometry, this point exists. The length of the perpendiculars drawn from point P to the X-axis and the Y-axis are both equal to 1.0 unit. However, the distance from the origin to the point where the perpendicular intersects the X-axis is equal to 0.7 units and is equal to the distance from the origin to the point of intersection of the perpendicular drawn from point P to the Y-axis. This example shows that although in Euclidian geometry, the statement that two parallel lines are equidistant is true; it is not necessarily true in hyperbolic geometry.

Until the twentieth century there was no practical use of the hyperbolic geometry. Modern technology and the discovery of the space showed that although our world is limited to three dimensions, existence of the pseudosphere is possible in the outer space. Our solar system was too small to show the distortion caused by the pseudospherical shape. However, scientists showed that hyperbolic geometry allowed achieving a closer approximation of Mercury's orbit than the approximation based on Euclidian geometry. These calculations proved that the sun causes some curvature. Since Mercury is closer to the Sun than Earth, it is located in a much higher gravitational field than our planet. Therefore, the space is significantly curved in Mercury's vicinity. In addition, with the help of hyperbolic geometry, physicists managed to come up with a theoretical explanation for the enigma of the black holes, and even though they still present a puzzling phenomenon, the assumption of four-dimensional space can shed some light on this mystery.

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Magic Squares: Going Beyond the Ordinary

By Chetan Mehta

Mathematics is not usually associated with recreation, but the fascinating topic of Magic Squares has historically been of interest to both mathematics enthusiasts and others casually interested in this relatively hip mathematical concept. While both parties are sufficiently knowledgeable on the nature of the subject, there are several interesting variations and sub-topics, unknown to most, which highlight the remarkable properties of these sophisticated mathematical objects.

The Basic Magic Square

A basic magic square consists of the positive integers $1, 2, \dots, n^2$ such that the sum of the n numbers in any horizontal, vertical, or *main* diagonal line is always the same Magic Constant (Weisstein 1999). Two examples are given below:

8	1	6
3	5	7
4	9	2

Magic Constant: 15

30	39	48	1	10	19	28
38	47	7	9	18	27	29
46	6	8	17	26	35	37
5	14	16	25	34	36	45
13	15	24	33	42	44	4
21	23	32	41	43	3	12
22	31	40	49	2	11	20

Magic Constant: 175

The magic constant is given by the equation:

$$M_2(n) = \frac{1}{n} \sum_{k=1}^{n^2} k = \frac{1}{2}n(n^2 + 1).$$

The Anti-Magic Square

An *Anti-Magic Square* is an arrangement of the numbers 1 to n^2 in a square matrix such that the row, column, and diagonal sums form a sequence of consecutive integers (Cormie 1999). An example is given below:

35	38	31	32	34
15	2	12	4	33
1	14	10	5	30
8	9	3	16	36
11	13	6	7	37
				39

There is no known method enabling us to construct Anti-Magic squares of order n . It has been proven that Anti-Magic squares of order 1 and 2 are impossible. In the case of order 1, the only possible arrangement is 1 which yields the same number for all values. In the

case of order 2, 6 consecutive sums are required, but only 5 are possible since the lowest obtainable value is $1 + 2 = 3$ and the highest is $3 + 4 = 7$. There are no known Anti-Magic Squares of order 3 as-of-yet and a compact proof excluding any possibilities of any such square is yet to be developed. There are several Anti-Magic Squares of order 4 and greater.

The Multiplication Magic Square

A normal multiplication magic square consists of the distinct positive integers such that the product of the n numbers in any horizontal, vertical, or *main* diagonal line is always the same magic constant. A very simple method of constructing a multiplication magic square is by raising any natural number a to every entry in a magic square of order n . For example ($a = 2, n = 4$):

Magic constant: 34

↓

Multiplication Magic Constant: 1717986918

4	14	15	1
9	7	6	12
5	11	10	8
16	2	3	13

16	16384	32768	2
512	128	64	4096
32	2048	1024	256
65536	4	8	8192

→

2^4	2^{14}	2^{15}	2^1
2^9	2^7	2^6	2^{12}
2^5	2^{11}	2^{10}	2^8
2^{16}	2^2	2^3	2^{13}

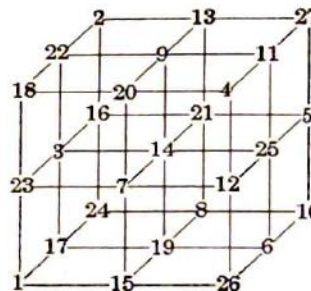
The Magic Cube and Magic Hexagon

A Magic Cube is essentially similar to a magic square but has 3 dimensions instead of 2, and contains the integers from 1 to n^3 . There are $3n^2 + 4$ lines that sum correctly. All rows, columns, pillars, and the four diagonals must sum to $1/2n(n^3+1)$ which is the magic constant. If the cross section diagonals sum to the magic constant, the magic cube is called a perfect magic cube; if they do not, the cube is called a semi-perfect magic cube (Hendricks 1999). Constructing a Magic Cube is very rigorous exercise, but an algorithm has been formulated by mathematician Marian Trenkler from Safarik University, Slovakia (check Works Cited for further information). An example is given below:

18	20	4
23	7	12
1	15	26

22	9	11
3	14	25
17	19	6

2	13	27
16	21	5
24	8	10



Magic Constant: 42

The Magic Hexagon is a unique case. Only two known Magic Hexagons, those of order 5 and order 1, exist. The diameter of a magic hexagon is always an odd number expressed

by $2n - 1$. In the magic hexagon below, each line (those of lengths 3, 4, and 5) adds up to 38. This is the only magic hexagon of the counting numbers for any size hexagon. It was discovered by C. W. Adams, who worked on this problem from 1910 to 1957. Mathematician C.W. Trigg showed that the magic constant for a hexagon of order n would be:

$$\frac{9(n^4 - 2n^3 + 2n^2 - n) + 2}{2(2n - 1)}$$

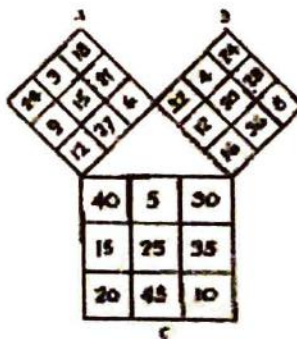
Which can be further simplified into $= (3n^2 - 3n + 1)(3n^2 - 3n + 2) / (4n - 2)$
 $= (1/32)(72n^3 - 108n^2 + 90n - 27 + 5/(2n-1))$

Thus requiring $5/(2n-1)$ to be an integer, possible only when $n = 1$ or 3 .



The Magic Triangle

A magic triangle is composed of three magic squares of the same size, but different magic constants. These figures extend the properties of magic squares to include the numeric properties of Pythagorean triples. As in the Magic Triangle shown below, the square of any cell in C is equal to the sum of the squares of the corresponding cells in A and B. For example: $25^2 = 15^2 + 20^2$. The square of the sum of any two or more cells or any row, column, or diagonal in C is equal to the square of the sum of the corresponding two or more cells, rows, columns, or diagonals of A, plus the square of the sum of the corresponding two or more cells, rows, columns, or diagonals of B (Houlton). For example: $(3+15+27)^2 + (12+20+28)^2 = (5+25+45)^2$.



In order to construct, set up three identical $n \times n$ magic squares with different magic constants. Select a Pythagorean triple (ex. (3,4,5), (5,12,13), (7, 24, 25)). Using the Pythagorean triple a, b, c , multiply each cell in square A by the value for a , each cell in

square B by the value for b, and each cell in square C by the value for c. Then rearrange the squares in a triangular form (Houlton).

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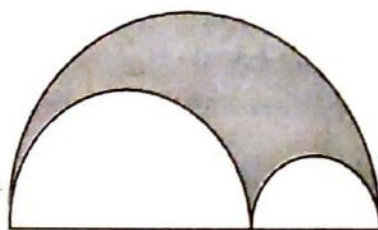
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Amazing Arbelos

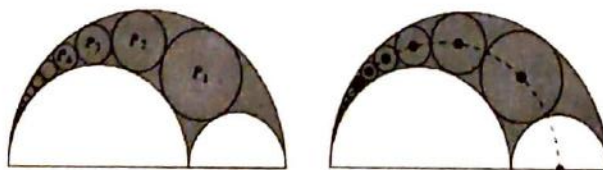
By Saish Setty



Archimedes, easily acknowledged as one of the greatest mathematicians in history, was intrigued by the arbelos. Named for its resemblance, arbelos was the Greek word for a shoemaker's knife. The arbelos is represented as the shaded portion of the above figure and possesses some very interesting characteristics.

Firstly, if one supposes that the diameter of the large semicircle is equal to one, then the diameters of the smaller semicircles are r and $1-r$, with $r < 1$. Then one knows that the combined arc lengths of the smaller semicircles would equal $\pi r + \pi(1-r)$ which equals π , the arc length of the largest semi-circle. Interestingly, the arc length of the smaller semicircles equals the arc length of the semicircle that encloses them.

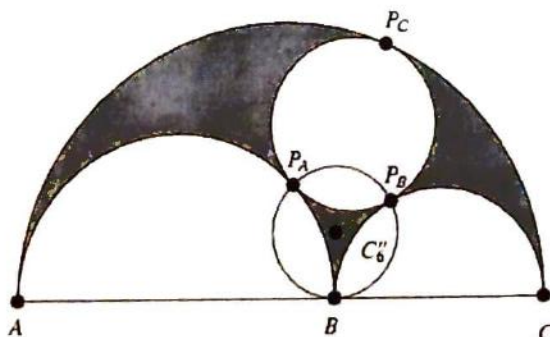
Another interesting property of the arbelos can be discovered if a perpendicular, BD is drawn from the tangent point of the two smaller semicircles to the edge of the largest semicircle. Fascinatingly, the area of the arbelos (the shaded region) is equal to the area of a circle with diameter BD . Numerous methods have been used to prove this property, including one involving the Pythagorean theorem. However, they are too lengthy to be accommodated within the confines of this article.



If a chain of tangent circles, a Pappus chain, beginning with a circle tangent to the small semicircles and the large semicircle, is constructed, then the centers of the circles lie on an ellipse. The diameter of the n^{th} circle, C_n , is equal to $(1/n)$ times the perpendicular distance to the base of the semicircle. This property was acknowledged by the mathematician Pappus, who referred to it as an ancient theorem. In addition, when the ratio of the diameter of the smaller semicircle to that of the smallest semicircle equals the golden ratio, then this chain of circles also fulfills a great many other properties.

One is also able to construct Archimedes' circles within the arbelos. If a perpendicular line is drawn from the point of tangency of the two smaller semicircles, one is able to construct two circles, tangent to the larger semicircle and this perpendicular. If one circle is also tangent to one of the smaller semicircles and the other circle is tangent to another smaller semicircle, then the two circles are congruent!

Another interesting circle within the arbelos is the Bankoff circle, which can be drawn using the first Pappus circle, P_1 . You may have noticed that P_1 is congruent to the Archimedes's circles. The Bankoff circle is tangent to point of tangency between the two smaller semicircles and the points at which the first Pappus circle is tangent to the smaller semicircles. The Bankoff circles' radius can then be represented as one half of the product of r and $1-r$.



The arbelos has enthralled many of the greatest mathematicians, including Descartes, Newton and Archimedes as each was intrigued by its amazing properties. Though this shape may have little use in our time, its astounding characteristics will continue to entice generations to come.

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Moments and the Moment-Generating Function (m.g.f.)

By Narani Pathmanapan

The s^{th} moment for the set of numbers $x_1, x_2, \dots, x_n$ is defined as

$$\overline{x^s} = \frac{x_1^s + x_2^s + \dots + x_n^s}{n} = \frac{\sum_{i=1}^n x_i^s}{n}$$

From this form, it is patent that the first moment is the arithmetic mean $\bar{x}$, because when $s=1$

$$\overline{x^1} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

The s^{th} moment about the mean $\bar{x}$ is defined as

$$m_s = \frac{\sum_{i=1}^n (x_i - \bar{x})^s}{n} = \overline{(x - \bar{x})^s}$$

The first moment about the mean is always equal to zero.

$$m_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})}{n} = \frac{\sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x}}{n} = \frac{\sum_{i=1}^n x_i}{n} - \frac{n\bar{x}}{n} = \bar{x} - \bar{x} = 0$$

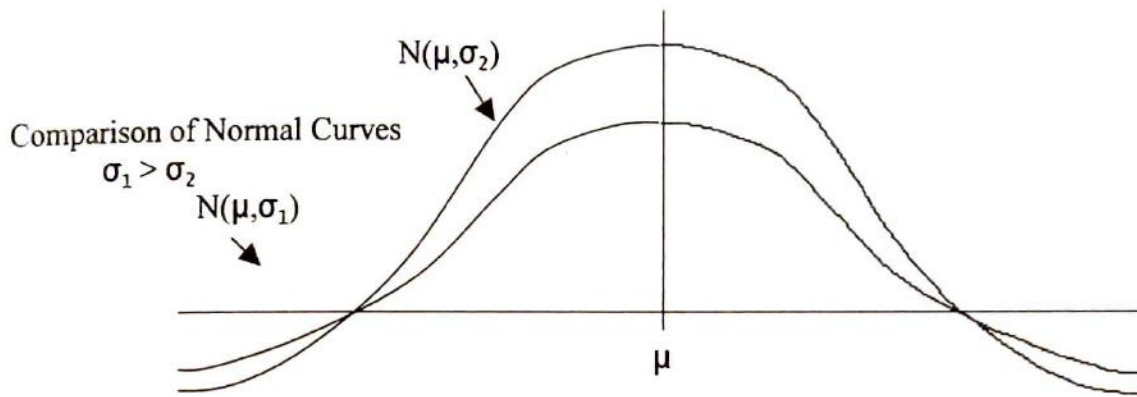
The second moment about the mean is variance σ^2 .

$$m_2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

Moments are used to describe distributions. The distribution of a variable tells us what values the variable takes and how often it takes these values, which helps us solve probability problems. The mean is the measure of the data's center and the variance is the measure of the data's spread.

There are many different types of distributions, but all of them fall into two categories: continuous and discrete. A continuous distribution notes all the values in some interval of numbers, for example height and weight. The normal distribution is one of the most important and commonly used continuous distributions and it is described with a normal curve.

Normal curves are part of a family called density curves. A density curve is always on or above the horizontal axis and has an area exactly 1 underneath the curve. Normal curves are symmetric, single peaked, and bell shaped.



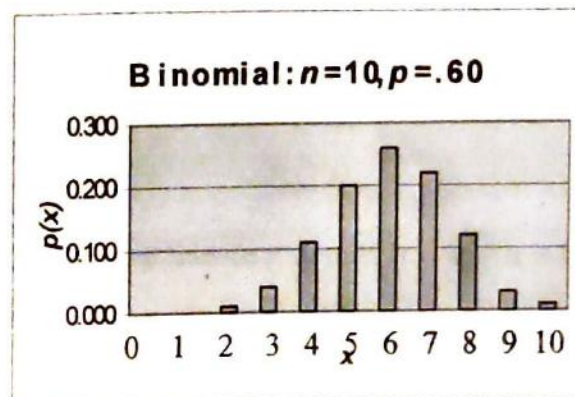
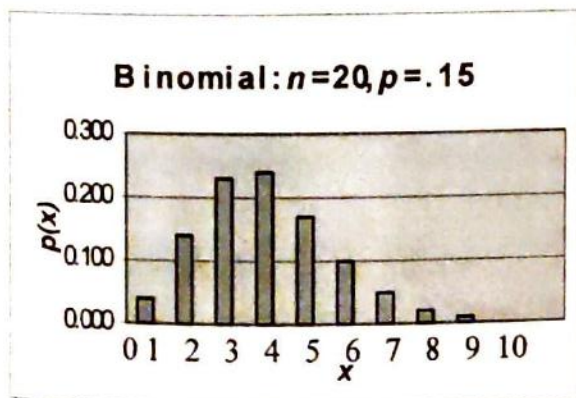
The normal density function is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \text{ for } -\infty < x < +\infty$$

By looking at the formula, one can see that normal distributions are defined by their mean μ and standard deviation σ . The probability of any event (its p.d.f.) is the area under the curve above the values that make up the event.

A discrete distribution notes a countable number of possible values. This is used for problems involving a fixed number of groups such as the sides on dice and coins. Under this banner falls the binomial distribution. It displays the probability of an event occurring. The parameters are n , the number of observations, and p , the probability of a success on any one observation. The binomial formula, where k is any value that X takes on, is

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \text{ for } k = 0, 1, 2, \dots, n$$



The probability distribution assigns each of these values a probability between 0 and 1 such that the sum of all the probabilities is exactly 1. The probability of any event is the sum of the probabilities of all the values that make up the event.

The moment-generating function for the normal distribution is

$$M_X(t) = e^{t\mu + \frac{\sigma^2 t^2}{2}}$$

The m.g.f. for the binomial distribution is

$$M_X(t) = [pe^t + (1-p)]^n$$

The moment generating function of both continuous and discrete distributions is just

$$M_X(t) = E(e^{tX}).$$

If X is a continuous random variable with p.d.f. f , we define the moment-generating function by

$$M_X(t) = \int_{-\infty}^{+\infty} e^{tx} f(x) dx$$

Proof of Normal m.g.f:

$$M_X(t) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\text{Let } \frac{x-\mu}{\sigma} = z; \text{ thus } x = \sigma z + \mu \text{ and } dx = \sigma dz$$

$$\therefore M_X(t) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{t(\sigma z + \mu)} e^{-\frac{z^2}{2}} dz$$

$$= e^{t\mu} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}[(z-\sigma t)^2 - \sigma^2 t^2]} dz$$

$$= e^{t\mu} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z^2 - 2\sigma t z)} dz$$

$$= e^{t\mu + \frac{\sigma^2 t^2}{2}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-\sigma t)^2} dz$$

Let $z - \sigma t = v$; then $dz = dv$ and we obtain

$$\begin{aligned} M_X(t) &= e^{t\mu + \frac{\sigma^2 t^2}{2}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{v^2}{2}} dv \\ &= e^{t\mu + \frac{\sigma^2 t^2}{2}} \end{aligned}$$

Let X be a discrete random variable with probability distribution $p(x_i) = P(X = x_i)$. Its m.g.f. is defined by

$$M_X(t) = \sum_{i=1}^{\infty} e^{tx_i} p(x_i)$$

Proof of Binomial m.g.f:

$$\begin{aligned}
 M_X(t) &= \sum_{k=0}^{\infty} e^{tx} \binom{n}{k} p^k (1-p)^{n-k} \\
 &= \sum_{k=0}^{\infty} \binom{n}{k} (pe^t)^k (1-p)^{n-k} \\
 &= [pe^t + (1-p)]^n
 \end{aligned}$$

M_X is called the moment generating function because it generates moments as can be seen below

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots \quad \text{Maclaurin series expansion of the function } e^x$$

$$e^{tx} = 1 + tx + \frac{(tx)^2}{2!} + \cdots + \frac{(tx)^n}{n!} + \cdots$$

$$M_X(t) = E(e^{tx}) = E\left(1 + tX + \frac{(tX)^2}{2!} + \cdots + \frac{(tX)^n}{n!} + \cdots\right)$$

$$M_X(t) = 1 + tE(x) + \frac{t^2 E(X^2)}{2!} + \cdots + \frac{t^n E(X^n)}{n!} + \cdots$$

Proof of Gauss's Integral Mean Value Theorem

By Winnie Tsao

We know from calculus that the average value of a function f on a closed interval $[a, b]$ is $\frac{1}{b-a} \int_a^b f(x) dx$. By substitution, considering a function of complex numbers, we

notice that the average value along a circle centered at z_0 is $\frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta$. In the

subsequent pages, we will prove that $\text{avg}_{z \in C} f(z) = f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta$, or that the average value along a circle is its center.

We begin with a sequence of **nested intervals**:

$$a_0 \leq a_1 \leq a_2 \leq \dots \leq a_n \leq \dots \leq b_n \leq \dots \leq b_2 \leq b_1 \leq b_0.$$

Since we know that every non-decreasing sequence bounded from above has a limit, we

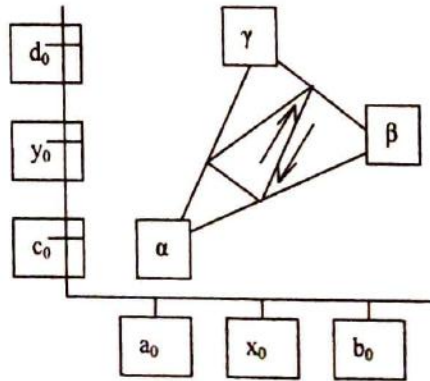
can say that for *this* sequence, $a_n \uparrow a$ and $b_n \downarrow b$, where the arrows indicate convergence.

Also, as $b_n - a_n \rightarrow 0$, the length of the entire interval goes to zero. So if $\lim_{n \rightarrow \infty} a_n - b_n = 0$,

then $a - b = 0$, and $a = b$. Knowing that $a_n \leq a$ and $b \leq b_n$, we can replace a and b with a

common value x_0 . Thus, $a_n \leq x_0 \leq b_n$.

Now, we construct a triangle with a medial triangle in it:



(The arrows indicate the orientation of the triangles.)

If $\int_C$ is the integral of $f(z)$ along the largest triangle, and $\int_{C_I}, \int_{C_{II}}, \int_{C_{III}}, \int_{C_{IV}}$ trace the paths along the four small triangles, then $\int_C = \int_{C_I} + \int_{C_{II}} + \int_{C_{III}} + \int_{C_{IV}}$, and $\left| \int_C \right| \leq \left| \int_{C_I} \right| + \left| \int_{C_{II}} \right| + \left| \int_{C_{III}} \right| + \left| \int_{C_{IV}} \right|$, by the **triangle inequality**.² Let C_I be that of $\{C_I, C_{II}, C_{III}, C_{IV}\}$ such that $\left| \int_{C_I} \right|$ is as large as possible. Thus, we can say $\left| \int_C \right| \leq 4 \left| \int_{C_I} \right|$. Similarly, we can also create a medial triangle inside C_I . If C_2 is one of the four triangles created such that $\left| \int_{C_2} \right|$ is greatest, then $\left| \int_{C_I} \right| \leq 4 \left| \int_{C_2} \right|$. Continuing this process, a pattern will arise:

$$\left| \int_C \right| \leq 4 \left| \int_{C_1} \right| \leq 4^2 \left| \int_{C_2} \right| \leq 4^3 \left| \int_{C_3} \right| \leq \dots \leq 4^n \left| \int_{C_n} \right|$$

If we project each of these triangles (C, C_I , etc.) onto the coordinate axes, we obtain two sequences of nested intervals, where the n^{th} interval is the projection of C_n . The points of accumulation of the two sequences are x_0 and y_0 . We can carry out this process for any triangular region T with boundary C lying in a **star-domain** D .³ So if we let $z_0 = x_0 + i y_0$, then $z_0 \in T$ and $f(z)$ is analytic at z_0 .⁴

We have $\frac{f(z) - f(z_0)}{z - z_0} = f'(z_0) + \frac{w(z)}{z - z_0}$, where $\frac{w(z)}{z - z_0}$ is a very small error

term, since we know that $f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$. Rearranging terms, we find that

¹ The triangle inequality states that $|z_1 + z_2| \leq |z_1| + |z_2|$, where $|z_1|$ and $|z_2|$ are the magnitudes of two vectors and $|z_1 + z_2|$ is the magnitude of the resultant vector. This can be stated more generally as $|\sum z_i| \leq \sum |z_i|$.

³ See appendix.

⁴ $f(z)$ is actually analytic on the entire star-domain. Since $T \subset D$ and $z_0 \in T$, we know that f is analytic at z_0 .

$f(z) = f(z_0) + f'(z_0)(z - z_0) + w(z)$, if z is near z_0 ($\lim_{z \rightarrow z_0}$). Since $f(z)$ is an analytic function, it can be expressed as $f(x + iy) = u(x, y) + iv(x, y)$, where $x, y \in \mathbb{R}$ and u, v are real functions. Given u is a differentiable function of (x, y) and f is differentiable, u and v are both differentiable. Then we have $u + iv = u_0 + iv_0 + [x + iy - x_0 - iy_0][u_x + iv_x] + w(z)$. This is also $u + iv = u_0 + iv_0 + [(x - x_0) + i(y - y_0)][u_x + iv_x] + w(z)$. Expanding, this is $u + iv = u_0 + iv_0 + [u_x(x - x_0) + iv_x(x - x_0) + iu_x(y - y_0) - v_x(y - y_0)] + w(z)$. We can now break this into its real and imaginary components. Looking first at the real part: $u = u_0 + u_x(x - x_0) - v_x(y - y_0) + p(x, y)$. By the **Cauchy-Riemann** equations, we have $u_y = -v_x$. Substituting, we get $u - u_0 = u_x(x - x_0) - u_y(y - y_0) + p(x, y)$, or simply $\Delta u = u_x \Delta x - u_y \Delta y + p(x, y)$. As $\Delta x, \Delta y \rightarrow 0$, $p(x, y) \rightarrow 0$. That means $\Delta u \rightarrow 0$. Now we can examine the imaginary part: $v = v_0 + v_x(x - x_0) - u_x(y - y_0) + q(x, y)$, where v is a differentiable function of (x, y) and given f is differentiable, u and v are both differentiable. Also by the **Cauchy-Riemann** equations, we have $u_x = v_y$. Similarly, we obtain $v - v_0 = v_x(x - x_0) - v_y(y - y_0) + q(x, y)$, or $\Delta v = v_x \Delta x - v_y \Delta y + q(x, y)$ through substitution. As $\Delta x, \Delta y \rightarrow 0$, $q(x, y) \rightarrow 0$. That means $\Delta v \rightarrow 0$. Since $w(z) = p + qi$, $w(z) \rightarrow 0$.

Thus we can express $f(z)$ as $f(z_0) + f'(z_0)(z - z_0) + w(z)(z - z_0)$, where

$|w(z)| < \varepsilon$, provided $|z - z_0| < \delta$. So we have:

$$\left| \int_{C_n} f(z) dz \right| \leq |f(z_0)| \left| \int_{C_n} dz \right| + |f'(z_0)| \left| \int_{C_n} (z - z_0) dz \right| + \left| \int_{C_n} w(z)(z - z_0) dz \right|$$

Looking at each term of the sum separately, we first realize that $\left| \int_{C_n} dz \right| = 0$ because C_n is a closed curve so its endpoints are equal, and $z|_{C_n} = 0$. The next term, $\left| \int_{C_n} (z - z_0) dz \right|$, is zero for the same reason (because $\frac{(z - z_0)^2}{2} \Big|_{C_n} = 0$). So we are left with $\left| \int_{C_n} f(z) dz \right| \leq \left| \int_{C_n} w(z)(z - z_0) dz \right|$. We know that for any function f and any curve γ , $\left| \int_{\gamma} f(z) dz \right| \leq \int_{\gamma} |f(z)| \cdot |dz|$. Thus, $\int_{C_n} |w(z)| |z - z_0| |dz| \leq |w(z)| \cdot |z - z_0| \cdot \int_{C_n} |dz|$ and $|w(z)| < \varepsilon$ and $|z - z_0| < \delta$. If we let s be the perimeter of triangle T , and let s_n be the perimeter of T_n , then $s_n = \frac{1}{2} s_{n-1} = \frac{1}{2^2} s_{n-2} = \dots = \frac{1}{2^n} s = \frac{s}{2^n}$, or $s_n^2 = \frac{s^2}{4^n}$. In fact, if $z \in T_n$, then $|z - z_0| \leq s_n$.⁵ Because $\int_{C_n} |dz| = s_n$, we have $\left| \int_{C_n} w(z)(z - z_0) dz \right| \leq \varepsilon \cdot s_n \cdot s_n$. Taken together, we have $\left| \int_C f(z) dz \right| \leq 4^n \left| \int_{C_n} f(z) dz \right| \leq 4^n \left(\frac{\varepsilon s^2}{4^n} \right)$, or $\left| \int_C f(z) dz \right| \leq \varepsilon s^2$. Because ε is arbitrarily close to zero and s^2 is a constant, εs^2 may be taken to be arbitrarily close to zero. $\left| \int_C f(z) dz \right|$ must be non-negative (due to the absolute value sign), so we have $\left| \int_C f(z) dz \right| = 0$. Since any complex number with a norm (or modulus) of zero is itself zero, $\int_C f(z) dz = 0$.

This can be generalized to all other, non-triangular curves. The **Cauchy-Goursat**

Theorem makes this generalization for all for star-domains:

Let D be a star-domain and let $f(z)$ be analytic in D . Let C be a closed curve in D . Then

$$\int_C f(z) dz = 0.$$

⁵ If $z \in T_n$, then z is a point on the triangle, z_0 is a point interior to the triangle, and s_n is the perimeter of the triangle. It can be observed that $|z - z_0|$, the length of the line segment

Having established this, we can go back to an equation we had earlier:

$$\frac{f(z) - f(z_0)}{z - z_0} = f'(z_0) + \frac{w(z)}{z - z_0}. \text{ This is equivalent to } \frac{f(z)}{z - z_0} - \frac{f(z_0)}{z - z_0} = f'(z_0) + \frac{w(z)}{z - z_0},$$

$$\text{or } \frac{f(z)}{z - z_0} = \frac{f(z_0)}{z - z_0} + f'(z_0) + \frac{w(z)}{z - z_0}. \text{ Integrating, we find that:}$$

$$\int_C \frac{f(z)}{z - z_0} dz = f(z_0) \int_C \frac{1}{z - z_0} dz + \int_C f'(z_0) dz + \int_C \frac{w(z)}{z - z_0} dz$$

Analyzing each term of the sum, we see that for the first term, $f(z_0) \int_C \frac{1}{z - z_0} dz =$

$$f(z_0) \int_K \frac{1}{z - z_0} dz. \text{ It follows that } f(z_0) \int_C \frac{1}{z - z_0} dz = f(z_0) \int_0^{2\pi} \left(\frac{1}{re^{i\theta}} \right) (rie^{i\theta} d\theta) =$$

$$f(z_0) \cdot i \int_0^{2\pi} d\theta = f(z_0) \cdot 2\pi i. \text{ For the second term, } \int_C f'(z_0) dz = f'(z_0) \int_C dz = f'(z_0) z \Big|_C$$

$$= f'(z_0) \cdot 0 = 0. \text{ For the third term, } \left| \int_C \frac{w(z)}{z - z_0} dz \right| = \left| \int_K \frac{w(z)}{z - z_0} dz \right|. \text{ Since}$$

$$\left| \frac{w(z)}{z - z_0} \right| = \frac{|w(z)|}{|z - z_0|} = \frac{|w(z)|}{r} \leq \frac{\varepsilon}{r} \text{ because } |w(z)| \leq \varepsilon, \quad \left| \int_K \frac{w(z)}{z - z_0} dz \right| \leq \frac{\varepsilon}{r} (2\pi r)$$

because we are integrating along the boundary of the circle, or the circumference.

$$\text{Thus } \left| \int_K \frac{w(z)}{z - z_0} dz \right| \leq 2\pi\varepsilon. \text{ We know that } \varepsilon \text{ is a very small positive number and that}$$

$2\pi\varepsilon$ can be any positive number with norm $< \varepsilon$. This forces $\left| \int_K \frac{w(z)}{z - z_0} dz \right|$ to be zero because it is nonnegative.

between z and z_0 , is shorter than the triangle's perimeter.

⁶ This is because $K: z_0 + re^{i\theta} = z$. Then $rie^{i\theta} d\theta = dz$, and since K avoids z_0 , $\frac{1}{z - z_0}$ is analytic on K .

Taken together, we have $\int_C \frac{f(z)}{z - z_0} dz = f(z_0) \cdot 2\pi i$. Rearranging, we get

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz.$$

What we have proven is the **Cauchy-Integral Theorem**: If $f(z)$

is analytic on star-domain D containing a circle C , center z_0 , radius r , and z_1 , is any point

on the interior of circle C , then $f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz$ where

$$C: z = z_1 + re^{i\theta} \ (0 \leq \theta \leq 2\pi), \quad z = C(t) = z_1 + re^{i\theta} \text{ and } dz = C'(t) dt = 0 + ire^{i\theta} d\theta.$$

Substituting for z and dz , we have $f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z_0 + re^{i\theta})}{re^{i\theta}} \cdot ire^{i\theta} d\theta$. Simplifying, we

arrive at the conclusion of our proof—indeed, $f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta$.

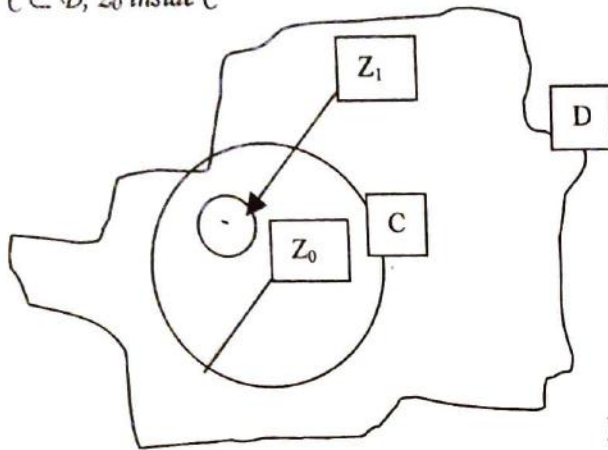
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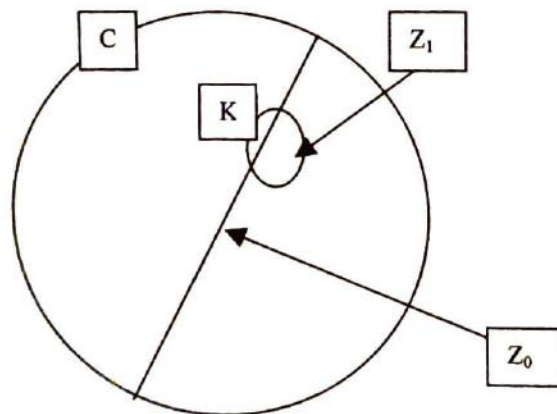
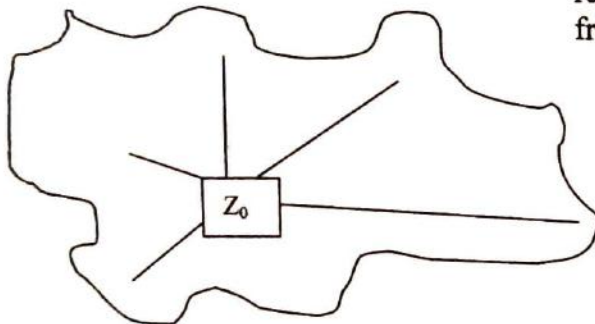
Appendix

C : a circle, centered at z_0 , radius r ,

$C \subset \mathcal{D}$, z_0 inside C



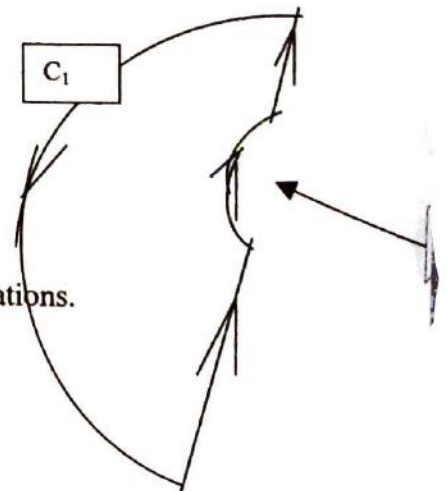
In a star-domain, every point on it, is reachable from a star-point (z_0) by a straight-line path.



C_1 and C_2 are two halves of the circle and have opposite orientations.
 $C_1 + C_2 = C - K$ because the opposite directions cancel out.

$$\begin{aligned} \int_{C_1} \frac{f(z)}{z - z_0} dz &= 0 \quad \left(\frac{f(z)}{z - z_0} \text{ avoids } z_0 \right) \\ + \int_{C_2} \frac{f(z)}{z - z_0} dz &= 0 \\ \hline \int_{C_1 + C_2} \frac{f(z)}{z - z_0} dz &= 0 \end{aligned}$$

$$\int_{C-K} = 0, \text{ so that } \int_C - \int_K = 0, \text{ or simply, } \int_C = \int_K$$



A Couple Public-Key Ciphers Other than RSA

By Andrew Laitman

Let's assume that there are 100 people in a room that want to communicate with each other person privately. Each person would thus have to have to maintain 99 keys and keep them private. Public key cryptography enables a party to receive information by publishing a key. Everyone knows how to encrypt a message in Public Key cryptography because the key is well, public. However, only one entity knows how to decrypt. In most algorithms, the decryption process can be derived from the encryption process, but in this case it is infeasible.

The security of the cipher is usually based on an intractable problem which is one that would take a computer a very long time to solve. For example, RSA encryption is based on how difficult it is to factor large numbers.

ElGamal public-key encryption

The security of this encryption scheme is based on the intractability of the discrete logarithm problem. Given the equation $x_1^y \equiv x_2 \pmod{p}$, x_1 , and x_2 , find the integer y . For example, $\log_5 35 \equiv 32 \pmod{97}$. No simple way of computing modular logarithms exists. In the case of mod 97, it would not take a long time for the computer to try every case. However, when a large prime is used (around 400 digits), it will take an extremely long time for a computer to try the correct case.

Summary of algorithm:

1. The receiver picks a large prime p and a generator α of the group $\mathbb{Z}_p^*$, which is the set of integers from 1 to $p-1$.
2. Choose an integer a , $1 \leq a \leq p-2$ and compute $\alpha^a \pmod{p}$. The reason why a cannot equal $p-1$ is because if it did α^{p-1} would equal 1 according to Fermat's Little Theorem.
3. The public key is the ordered triple (p, α, α^a) . The private key is a .

Algorithm:

1. Encryption:
 - (a) Obtain the public key (p, α, α^a) .
 - (b) Transform the message into an integer m in the group $\mathbb{Z}_p^*$. For example, one can represent the letter 'D' as 04.
 - (c) Choose an integer k , $1 \leq k \leq p-2$.
 - (d) Compute $\gamma = \alpha^k \pmod{p}$ and $\delta = m(\alpha^a)^k \pmod{p}$.
 - (e) Send $c = (\gamma, \delta)$.
2. Decryption

(a) Since the receiver has a he or she can compute

$$\gamma^{p-1-a} \pmod{p} \quad (\gamma^{p-1-a} \equiv \gamma^{-a} \equiv \alpha^{-ak}).$$

(b) Find m by computing $(\gamma^{-a})\delta \pmod{p}$.

Proof: $\gamma^{-a}\delta \equiv \alpha^{-ak}m\alpha^{ak} \equiv m \pmod{p}$.

Goldwasser – Micali Public Key Encryption

The Goldwasser – Micali encryption scheme is based on the difficulty of the quadratic residuosity problem. Before going over the scheme, I need to describe Legendre and Jacobi symbols.

Legendre and Jacobi symbols determine whether or not a number is a quadratic residue in a certain mod. A quadratic residue, r , exists when there is a solution to the equation $x^2 \equiv r \pmod{p}$. Only half of the numbers in the group $\mathbb{Z}_p^*$ are quadratic residues.

How to determine if r is a quadratic residue $\pmod{p}$:

Let Q_p be the set of quadratic residues $\pmod{p}$. The Legendre symbol has 3 possible values, where p is an odd prime and a , an integer.

$$\left(\frac{a}{p}\right) = 0, \text{ if } p \text{ divides } a$$

$$\left(\frac{a}{p}\right) = 1, \text{ if } a \in Q_p$$

$$\left(\frac{a}{p}\right) = -1, \text{ if } a \notin Q_p$$

The Jacobi symbol is very similar to the Legendre symbol, except that the denominator can be any positive integer greater than 3. A Jacobi symbol can be divided into a Legendre symbol into Legendre symbols in the following manner:

$$\left(\frac{a}{p_1 p_2}\right) = \left(\frac{a}{p_1}\right) \left(\frac{a}{p_2}\right)$$

The Goldwasser-Micali encryption scheme is based on the ability to compute the value of Jacobi symbols. That problem is based on the difficulty of factoring large numbers into primes.

Algorithm for key generation:

1. Choose two large and distinct primes p and q .
2. Compute $n = pq$.
3. Choose a nonzero integer y that is less than n such that y is a quadratic non-residue.
4. The public key is (n, y) .

Algorithm:**1. Encryption**

- (a) Obtain the public key (n, y) .
- (b) Write the message as a binary string $m = m_1 m_2 \dots m_t$ of length t .
- (c) For each i from 1 to t :
 - i. Choose an x that is a member of the group $\mathbb{Z}_p^*$ randomly.
 - ii. Set $c_i = yx^2$ if $m_i = 1$. Set $c_i = x^2$ if $m_i = 0$.
- (d) Send the sequence $c = (c_1 c_2 \dots c_t)$ to the receiver.

2. Decryption

- (a) For each i from 1 to t :
 - i. Determine the value of the Legendre symbol $e_i = \left(\frac{c_i}{p}\right)$.
 - ii. If $e_i = 1$ then m_i must be 0 and vice versa.
- (b) $m = m_1 m_2 \dots m_t$

Explanation:

If m_i is equal to 0, then $c_i = x^2$ is a quadratic residue mod n . If $m_i = 1$, then vice versa. For example if I have 5^2 , 2×5^2 cannot possibly be a square. The only way to break this code, or by determining if a number is a perfect square is by knowing how to compute the Legendre symbol. It is impossible to compute the symbol without knowing how to factor n .

Optimal Selection of a Service Center

By Xiru (Luke) Li

1. Introduction

A service center is a place or location that provides service to people in the surrounding areas. Examples of service centers include parking lots, warehouses, and grocery stores. In the airport, a parking lot is shared by several airline terminals. A deli store serves people in its vicinity. A fire department usually covers regions within a certain distance. Now a natural question arises- where should the service center be located? It would be fair if people from various terminals would take on average the least amount of time to reach the shared parking lot. It is important for a fire engine to reach the regions it covers in the shortest amount of time. We could name other examples to illustrate the importance of the location of a service center.

All these arguments relate to the optimal design of a service center. This design usually depends on so-called optimization theory. Optimization is a process to model a real life problem by setting an objective function and possibly some constraints, then finding a feasible maximum or minimum solution to the objective function that is within the constraints. The key to optimization is finding the solution. Among many studies, the most important contribution is done by the mathematicians Kuhn and Tucker, and is called Kuhn-Tucker conditions. This is the backbone of optimization.

We will first discuss the service center design under no constraints in Section 2. In Section 3, under constraints, a service center is modeled. Its solution is derived with the help of the Kuhn-Tucker theorem.

2. Optimal Service Center Selection With No Constraint

Suppose there is a service center that is going to serve several locations near it. Apparently, an ideal service center should save the time people travel to it. In finding the optimized solution, a criterion would be that the total distance from each location (that the center serves) to the service center is minimized. It is intuitive that the service center shared by two buildings should be on the midpoint of the line segment that connects these two buildings. In general, let us assume that on a coordinate plane the location of a service center is (x, y) and the locations to be served are $(a_1, b_1), (a_2, b_2), \dots, (a_n, b_n)$ where a_i, b_i could be positive or negative for $i = 1, 2, \dots, n$ and n is the total number of locations to be served. For analytical tractability, as a standard mathematical approach, we set our goal to minimize the sum of the squares of the distances from the locations to be served to the service center. That is, to minimize the quantity $\sum_{i=1}^n [(a_i - x)^2 + (b_i - y)^2]$. The optimization problem can be modeled as

$$\text{Minimize} \quad \sum_{i=1}^n [(a_i - x)^2 + (b_i - y)^2] \quad (2.1)$$

This is called unconstrained optimization. The value of (x, y) that solves (2.1) would be the coordinate of the location of our optimal service center.

Note, we could also choose the quantity to be $\sum_{i=1}^n \sqrt{(a_i - x)^2 + (b_i - y)^2}$, which is the sum of the distances from the locations to be served to the service center. However, we won't be able to solve it analytically. For showing the application of optimization, we picked the quantity without the square root. It will be our future research to solve the square root objective function numerically.

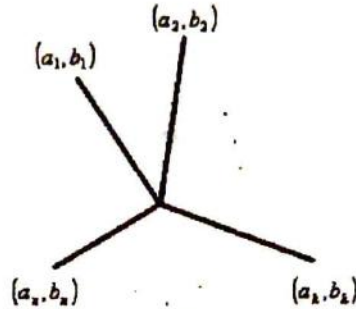


Figure 2.1

The following Theorem 2.1 shows us how to solve the unconstrained problem (2.1)

Theorem 2.1 The solution to the problem (2.1) is given by

$$\left(x = \frac{1}{n} \sum_{i=1}^n a_i, y = \frac{1}{n} \sum_{i=1}^n b_i \right) \quad (2.2)$$

Proof The first order condition for a differentiable two-variable function gives us

$$\sum_{i=1}^n 2(a_i - x) = 0, \sum_{i=1}^n 2(b_i - y) = 0 \text{ or } \left(x = \frac{1}{n} \sum_{i=1}^n a_i, y = \frac{1}{n} \sum_{i=1}^n b_i \right).$$

By the nature of the problem, we know the optimal solution, which is a minimum, exists. Further, the optimal solution should satisfy the first order condition. Since (2.2) is the unique solution to the first order condition, we conclude that (2.2) is the required optimal solution to the problem.

Note, here we avoided introducing the second order condition which addresses the sufficiency of a solution derived from the first order condition. Its definition can be found in the References. Interested readers can verify that solution (2.2) does satisfy the second order condition. The optimality can also be verified as a special case of Theorem 3.2. QED

Solution (2.2) is actually the algebraic average of the coordinates of these n points. Let us exam some special cases with the help of Theorem 2.1. The solution to a two-location served, one service center problem is given by $\left(x = \frac{a_1 + a_2}{2}, y = \frac{b_1 + b_2}{2} \right)$,

which agrees with our intuition mentioned above. A more interesting example is a three-location, one service center problem. It will have the solution

$$\left(x = \frac{a_1 + a_2 + a_3}{3}, y = \frac{b_1 + b_2 + b_3}{3} \right).$$

For a triangle with $(a_1, b_1), (a_2, b_2), (a_3, b_3)$ being its three endpoints, the solution is the center of balance of the triangle or the centroid of the triangle (the point where the three medians intersect).

People will ask if the service center cannot be constructed in certain areas because of the limit of the available land, conflict with the existing construction and so on. In these cases, the solution (2.2) might not give a feasible optimum; i.e., the algebraic average point might be in the area that is practically forbidden to get in. Those difficulties or practical impossibilities for a solution can happen in optimization.

In optimization, (2.1) is called an unconstrained problem. It means the solution is not subjected to any constraint or a physical limit in practice. Usually, it is not very common to see the unconstrained optimization in real life. More or less, limits or constraints are imposed on the problem. To address the optimal service center problem in practice, we will need to adopt a new methodology, which brings us to our next section.

3. Optimal Service Center Selection With Constraints

Generally, there are all sorts of constraints which could be imposed on the service center selection. For example, a constraint on the distance of building A to the service center can result if the building is a retirement home for elders. Unlike the case in Section 2, the service center itself can be prohibited from being within a set radius of some location.

Since there are various constraints to the problem and one cannot discuss all of them, we are going to focus on the problem with distance constraint. The objective function is still the same as in Section 2, which is to minimize the sum of the squares of the distances. Let c be the maximum allowed distance from a building k to the service center. Then, the distance constraint takes the form

$$(a_k - x)^2 + (b_k - y)^2 \leq c^2.$$

The optimal service center selection problem can be expressed as

$$\text{Minimize } \sum_{i=1}^n [(a_i - x)^2 + (b_i - y)^2] \quad (3.1)$$

$$\text{Subject to } (a_k - x)^2 + (b_k - y)^2 \leq c^2$$

Here we only assume one constraint for ease of illustration.

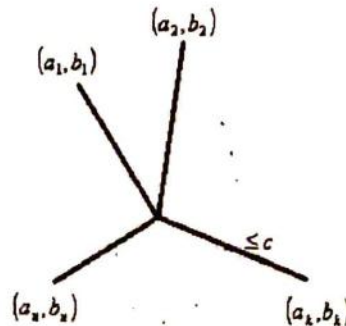


Figure 3.1

Before solving the problem, we need to introduce the convex property for a multi-variable function. It is known that for a single variable, twice differentiable function $f(x)$, it is convex if and only if $f_{xx} \geq 0$. For a two-variable, twice differentiable function $f(x, y)$, it is convex if and only if $f_{xx} \geq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$. A related constrained convex optimization problem is defined as

$$\begin{aligned} & \text{Minimize } f(x, y) \\ & \text{Subject to } g(x, y) \leq 0 \end{aligned} \quad (3.2)$$

where both $f(x, y)$ and $g(x, y)$ are twice differentiable and convex.

It is intuitive that a convex function has a minimum point. Let us state without proof the famous Kuhn-Tucker theorem for this constrained convex optimization problem.

Theorem(Kuhn-Tucker) 3.1 Assume functions $f(x, y)$ and $g(x, y)$ are twice differentiable and convex. If there exist (x^*, y^*) and λ^* , such that

$$f_x(x^*, y^*) + \lambda^* g_x(x^*, y^*) = 0, \quad (3.3)$$

$$f_y(x^*, y^*) + \lambda^* g_y(x^*, y^*) = 0 \quad (3.4)$$

$$\lambda^* g(x^*, y^*) = 0, \quad (3.5)$$

$$\lambda^* \geq 0,$$

$$g(x^*, y^*) \leq 0.$$

Then, (x^*, y^*) is the optimal solution to problem (3.2).

Let $f(x, y) = \sum_{i=1}^n [(a_i - x)^2 + (b_i - y)^2]$, $g(x, y) = (a_k - x)^2 + (b_k - y)^2 - c^2$. The

corresponding derivatives are given by:

$$f_{xx} = 2n > 0, f_{yy} = 2n, f_{xy} = 0, f_{xx}f_{yy} - f_{xy}^2 = 4n^2 > 0.$$

$$g_{xx} = 2 > 0, g_{yy} = 2, g_{xy} = 0, g_{xx}g_{yy} - g_{xy}^2 = 4 > 0.$$

Therefore, f and g are convex functions and so our problem (3.1) is a constrained convex optimization. We need to solve the equations in the Kuhn-Tucker theorem. The following theorem gives us the optimal solution and the procedure to solve the problem.

Theorem 3.2 If $\left(a_k - \frac{1}{n} \sum_{i=1}^n a_i\right)^2 + \left(b_k - \frac{1}{n} \sum_{i=1}^n b_i\right)^2 > c^2$, the optimal solution is given by

$$\begin{aligned} (x^*, y^*) &= \left(\frac{\sum_{i=1}^n a_i + \lambda^* a_k}{n + \lambda^*}, \frac{\sum_{i=1}^n b_i + \lambda^* b_k}{n + \lambda^*} \right), \\ \lambda^* &= \frac{n}{c} \sqrt{\left(a_k - \frac{1}{n} \sum_{i=1}^n a_i\right)^2 + \left(b_k - \frac{1}{n} \sum_{i=1}^n b_i\right)^2} - n > 0. \end{aligned}$$

Otherwise, if $\left(a_k - \frac{1}{n} \sum_{i=1}^n a_i\right)^2 + \left(b_k - \frac{1}{n} \sum_{i=1}^n b_i\right)^2 \leq c^2$, then

$$(x^*, y^*) = \left(\frac{\sum_{i=1}^n a_i}{n}, \frac{\sum_{i=1}^n b_i}{n} \right),$$

$$\lambda^* = 0.$$

where (x^*, y^*) is the optimum location for our service center.

Proof: First let us solve equation (3.3) and equation (3.4) by plugging in the first order derivatives f_x, f_y and g_x, g_y , we get

$$\sum_{i=1}^n (a_i - x) + \lambda(a_k - x) = 0, \sum_{i=1}^n (b_i - y) + \lambda(b_k - y) = 0.$$

or

$$\sum_{i=1}^n a_i - nx + \lambda a_k - \lambda x = 0, \sum_{i=1}^n b_i - ny + \lambda b_k - \lambda y = 0.$$

Rearrange terms to solve for x and y ,

$$(x^*, y^*) = \left(\frac{\sum_{i=1}^n a_i + \lambda a_k}{n + \lambda}, \frac{\sum_{i=1}^n b_i + \lambda b_k}{n + \lambda} \right).$$

Now we are ready to prove part 1 under $\left(a_k - \frac{1}{n} \sum_{i=1}^n a_i\right)^2 + \left(b_k - \frac{1}{n} \sum_{i=1}^n b_i\right)^2 > c^2$.

From equation (3.5), if $\lambda > 0$, then $(a_k - x)^2 + (b_k - y)^2 - c^2 = 0$. Replace x and y by x^* and y^* from above, we have

$$\left(\frac{na_k - \sum_{i=1}^n a_i}{n + \lambda} \right)^2 + \left(\frac{nb_k - \sum_{i=1}^n b_i}{n + \lambda} \right)^2 - c^2 = 0,$$

which leads to the solution for λ ,

$$\lambda^* = \frac{n}{c} \sqrt{\left(a_k - \frac{1}{n} \sum_{i=1}^n a_i\right)^2 + \left(b_k - \frac{1}{n} \sum_{i=1}^n b_i\right)^2} - n.$$

There is no difficulty in verifying that if $\left(a_k - \frac{1}{n} \sum_{i=1}^n a_i\right)^2 + \left(b_k - \frac{1}{n} \sum_{i=1}^n b_i\right)^2 > c^2$, then

$\lambda^* > 0$ is true. Furthermore, $(a_k - x^*)^2 + (b_k - y^*)^2 - c^2 = 0$ holds true, or the constraint

is satisfied. In this way, we solved the problem and verified that (x^*, y^*) and λ^* satisfy all the Kuhn-Tucker conditions.

The second part under $\left(a_k - \frac{1}{n} \sum_{i=1}^n a_i\right)^2 + \left(b_k - \frac{1}{n} \sum_{i=1}^n b_i\right)^2 \leq c^2$ is straightforward.

As an exercise, readers can verify that $(x^*, y^*) = \left(\frac{\sum_{i=1}^n a_i}{n}, \frac{\sum_{i=1}^n b_i}{n}\right)$ and $\lambda^* = 0$ satisfy Kuhn-

Tucker conditions. QED.

The theorem shows us how to solve the problem and the solution relationship between the unconstrained problem (2.1) and the constrained problem (3.1). We could first solve the unconstrained problem (2.1) with the solution $\left(x = \frac{1}{n} \sum_{i=1}^n a_i, y = \frac{1}{n} \sum_{i=1}^n b_i\right)$. Then check whether the constraint holds true with the solution. If it is true, the unconstrained and constrained problems share the same solution. Otherwise, we solve λ^* and adjust the unconstrained solution by λ^* to get the constrained solution following the theorem. The constrained optimization problem here is a generalization of the problem without constraints in Section 2.

4. Conclusion

We introduced optimization theory to resolve the service-center locating problem. The solutions to both unconstrained and constrained problems are given analytically. The unconstrained problem solution has a natural interpretation. In general, the constrained problem is much difficult to solve. For illustrative purposes, we only touched one distance constraint case. The real problems could have multiple distance constraints. Then the optimal solution will reside in the intersection of all those circles formed by the constraints. In real life, the constraint might not necessary take the form of distance we discussed, and the problem might not even be convex. In addition, as we previously noted, the original objective function involves square root that are sometimes difficult to solve analytically. How to solve them is a challenge that will require us to explore not only analytical solutions but numerical solutions as well.

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The Beauty of Mathematics

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The long tradition of Stuyvesant High School sending gifted (and obviously well-prepared) mathematics students out into the world is legend. At the same time we find ourselves in a society where the average person takes pride in having been weak in mathematics in school and enjoys telling about his continuous ignorance of the subject. Yet when these Stuyvesant graduates go among the rest of the populace, they have an obligation of spreading the word about the beauty of mathematics. Armed with the appropriate enthusiasm, they should begin to develop an arsenal of simple illustrations in the basic areas of mathematics that demonstrate this "beauty." Stuyvesant has a long list of Math Department heads who have supported this theme. Some of the more recent ones are Benjamin Bold, Henrietta Midonick, Edward Marcantonio, and Richard Rothenberg. Key among them is the current assistant principal for mathematics, Daniel

$$\begin{array}{rclcl}
 1 & \bullet & 8 & + & 1 & = & 9 \\
 12 & \bullet & 8 & + & 2 & = & 98 \\
 123 & \bullet & 8 & + & 3 & = & 987 \\
 1,234 & \bullet & 8 & + & 4 & = & 9,876 \\
 12,345 & \bullet & 8 & + & 5 & = & 98,765 \\
 123,456 & \bullet & 8 & + & 6 & = & 987,654 \\
 1,234,567 & \bullet & 8 & + & 7 & = & 9,876,543 \\
 12,345,678 & \bullet & 8 & + & 8 & = & 98,765,432 \\
 123,456,789 & \bullet & 8 & + & 9 & = & 987,654,321
 \end{array}$$

Jaye, a former outstanding student of mine at CCNY, (class of 1972), who has done much within the school to "infect" students with the beauty of mathematics and beyond that has forged a strong relationship with his alma mater. Some of his creative ideas that have borne fruit include a summer math academy at CCNY, and he has many Stuyvesant students taking CCNY courses before their high school graduation.

So now to provide some motivation and to offer some little reminders about the beauty of mathematics, a few short illustrations are provided here for your entertainment.

To begin notice the beauty in number patterns such as the following.

Then there are the beautiful internal number patterns that exist for some special numbers. For example, if we find the square of the sum of the digits of 81 we get:

$$81 = (8 + 1)^2 = 9^2$$

Now consider a three-digit number where this astonishing relationship holds:

$512 = (5 + 1 + 2)^3 = 512$. Clearly, this does not happen for all numbers. Since there are no

other two-digit numbers or three-digit numbers for which this works, here are the four-digit numbers and some five-digit numbers where this is true:

$$\begin{aligned} 4,913 &= 17^3 \\ 5,832 &= 18^3 \\ 17,576 &= 26^3 \\ 19,683 &= 27^3 \end{aligned}$$

Perhaps you are ambitious and would like to solve a heretofore-unsolved problem. See if you can prove that the following will always be true:

Select any natural number.

- If the number is odd then multiply by 3 and add 1.
- If the number is even then divide by 2.

Regardless of the number you select, you will always end up with 1.

Regardless of whether or not you are able to prove this (remember, it has never been proved), it is fun to see it always work.

How about showing someone a pair of *friendly numbers*. These are a pair of numbers where the sum of the proper factors of one number equals the second number *and* the sum of the proper factors of the second number equals the first number as well. Consider the smallest pair of friendly numbers: 220 and 284.

The divisors (or factors) of 220 are 1, 2, 4, 5, 10, 11, 20, 22, 44, 55, and 110. Their sum is $1+2+4+5+10+11+20+22+44+55+110 = 284$.

The divisors of 284 are 1, 2, 4, 71, and 142, and their sum is $1+2+4+71+142 = 220$.

The second pair of friendly numbers (discovered by Pierre de Fermat) is: 17,296 and 18,416.

$$17,296 = 2^4 \cdot 23 \cdot 47, \text{ and } 18,416 = 2^4 \cdot 1151.$$

The sum of the factors of 17,296 is

$$1 + 2 + 4 + 8 + 16 + 23 + 46 + 47 + 92 + 94 + 184 + 188 + 368 + 376 + 752 + 1081 + 2162 + 4324 + 8648 = 18,416$$

The sum of the factors of 18,416 is

$$1 + 2 + 4 + 8 + 16 + 1151 + 2302 + 4604 + 9208 = 17,296$$

There are many more pairs of friendly numbers.

Some problem-solving situations require some "out of the box" thinking that leaves you with some favorable lasting effects. Let's consider the question: Where on earth can you be so that you can walk one mile south, then one mile east, and then one mile north and end up at the starting point?

Yes, the North Pole. To test this answer, try starting from the North Pole and traveling south one mile and then east one mile. This takes you along a latitudinal line, which remains equidistant from the North Pole, one mile from it. Then travel one mile north to get you back to where you began, the North Pole.

Most people familiar with this problem feel a sense of completion. Yet we can ask: Are there other such starting points, where we can take the same three “walks” and end up at the starting point? The answer, surprising enough for most people, is *yes*.

One set of starting points is found by locating the latitudinal circle, which has a circumference of one mile and is nearest the South Pole. From this circle, walk one mile north (along a great circle, naturally), and form another latitudinal circle. Any point along this second latitudinal circle will qualify. Let’s try it.

Begin on this second latitudinal circle (the one farther north). Walk one mile south (takes you to the first latitudinal circle), then one mile east (takes you exactly once around the circle), and then one mile north (takes you back to the starting point).

Suppose the first latitudinal circle, the one we would walk along, would have a circumference of $\frac{1}{2}$ mile. We could still satisfy the given instructions, yet this time walking around the circle *twice*, and get back to our original starting point. If the first latitudinal circle had a circumference of $\frac{1}{4}$ mile, then we would merely have to walk around this circle *four* times to get back to the starting point of this circle and then go north one mile to the original starting point.

At this point we can take a giant leap to a generalization that will lead us to many more points that satisfy the original stipulations, actually an infinite number of points! This set of points can be located by beginning with the latitudinal circle, located nearest the south pole, which has a $\frac{1}{n}$ -mile circumference, so that a 1-mile walk east (which comprises n circumnavigations) will take you back to the point on the circle at which you began your walk on this latitudinal circle. The rest is the same as before, that is walking one mile south and then later one mile north. Is this possible with latitude circle routes near the North Pole? Yes, of course!

There are many other illustrations that exhibit the beauty of mathematics. In the field of probability, the “birthday problem” shows –against all intuition – that it is practically certain that two people in a group of 55 will have the same birth date. Probability also tells us that the 13th of a month is more likely to be on a Friday than any other day of the week. There are wonderfully demonstrable invariants in geometry that further show mathematics’ beauty for what it is – highly motivational. Stuyvesant graduates must show the way to these treasures in a way that is contagious to others.

How vs. Why: A Quest for Moving Curtains

Paul Zeitz, Class of 1975

1 Introduction: Two Familiar Arguments

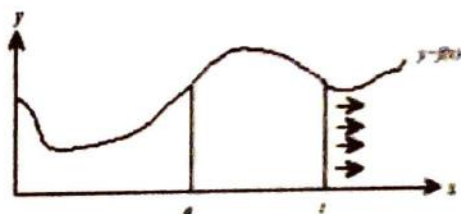
Many proofs are logically sound, in that each step follows from the previous one, yet the proof as a whole sheds very little light on whatever was proven. We call such proofs *Hows*, in contrast to *Whys*, which are proofs that dramatically get to the heart of the matter. The phrase "Moving Curtain" can be used interchangeably with *Why*, in honor of the following quick demonstration of the Fundamental Theorem of Calculus:

We wish to show that

$$\int_a^b f(x) dx = F(b) - F(a).$$

where F is any antiderivative of f ; i.e., $F'(x) = f(x)$. Define

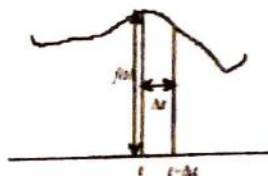
$$g(t) := \int_a^t f(x) dx.$$



The function $g(t)$ thus measures the area of a "moving curtain" (the shaded area above). It makes intuitive sense that

$$g'(t) = f(t).$$

since in a small interval of time Δt , the curtain's area will grow by approximately $f(t) \Delta t$.



We deliberately called this a demonstration, not a proof, since the argument as it stands is not at all rigorous. Lots of technical details are needed before we can even call it correct, with confidence.

But that's not the point. The Moving Curtain argument tells us elegantly, economically, and emphatically, *why* the Fundamental Theorem of Calculus is true. And these are features of most Moving Curtain arguments. They make creative use of "natural" points of view (in the above example, a dynamical picture), and the rest seems almost inevitable. After you absorb a *Why*, you feel that you truly

understand, and should be able to explain it in a few words: *The rate of change of the area of a unit-velocity moving curtain is equal to the height of the leading edge of the curtain.*

Here is an equally well-known How argument: a proof of Euler's formula

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Using power series, we have

$$\begin{aligned} e^{i\theta} &= \sum_{k=0}^{\infty} \frac{(i\theta)^k}{k!} \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k}}{(2k)!} + i \left(\sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k+1}}{(2k+1)!} \right) \\ &= \cos \theta + i \sin \theta. \end{aligned}$$

Of course, this is no more rigorous than the previous argument; many questions have to be addressed before it even makes sense. Nevertheless, it is the heart of a rigorous and important argument, and a beautiful one, as well. We are sure that the reader was deeply impressed by this argument when he or she saw it for the first time.

Yet it is not a Why. It sheds no light on why the formula is true. Using power series makes the trig functions appear, almost magically. Magic can be beautiful and inspiring, but it rarely illuminates. In other words, the set of Whys may have a large intersection with Erdos's "Book," but the two sets are not equal.

This article will discuss a number of Whys, contrasting them with Hows, in several branches of (mostly elementary) mathematics. We hope that it spurs the reader to find many more Whys. Of course we are not advocating the elimination of Hows in mathematics, but we do contend that true understanding eludes one in the absence of a Why.

2 Number Theory

Products of Consecutive Integers

We shall begin with a very simple problem, one of the first exercises encountered when studying mathematical induction:

Prove that the product of three consecutive integers is divisible by 6.

Certainly, induction works just fine here, and gives us confidence to generalize to products of four or more consecutive integers. It is natural to conjecture that

The product of r consecutive integers is divisible by $r!$.

But a better argument avoids induction altogether—induction rarely leads to Whys—and observes that at least one of n , $n+1$, $n+2$ must be even and exactly one must be a multiple of 3. Hence (since 2 and 3 are relatively prime) the product is a multiple of 6. This method looks generalizable, but with larger products, say 28 consecutive integers, some difficulties arise. What do you do when

factors are not relatively prime?

What we shall do is change our point of view from number theory to combinatorics. This is not a strange thing to do; any time integers are involved in a problem, it may be the case that simple counting lies at its heart. And indeed, products of consecutive integers are permutations! More precisely, a product of r consecutive integers can be written as

$$n(n-1)(n-2)\cdots(n-r+1).$$

for some integer n , and this is equal to the number of ways of permuting r objects chosen from a set of n objects. And surely this is a multiple of $r!$, since

$$\frac{n(n-1)(n-2)\cdots(n-r+1)}{r!} = \binom{n}{r}.$$

the number of ways of choosing r *unordered* objects from a set of n objects. In other words, the fact that $\binom{n}{r}$ is an integer is *why* $r!$ must divide the product of r consecutive integers!

Fermat's Little Theorem

Sometimes several different Whys explain the same problem. Fermat's Little Theorem is an example that has two nice Whys: one relatively low-tech, the other slightly high-tech.

We wish to prove that if p is a prime, then $a^p \equiv a \pmod{p}$ for any integer a . The classical proof notes that $a \equiv 0 \pmod{p}$ is trivial; eliminating this case allows us to equate two sets (modulo p):

$$\{a, 2a, 3a, \dots, (p-a)a\} \equiv \{1, 2, 3, \dots, p-1\} \pmod{p}.$$

This follows because p is a prime; it is easy to show that the elements of the left-hand set are distinct and non-zero (modulo p).

Equating the products yields

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$$

Because p is prime, $(p-1)$ is nonzero (modulo p), so we can "cancel," getting

$$a^{p-1} \equiv 1 \pmod{p}.$$

the non-trivial case of Fermat's Little Theorem.

This is a nice proof, but it is not quite a Why. We can turn it into a Why with a single sentence: All of what we did worked because $\{1, 2, 3, \dots, p-1\}$ is a *group* under multiplication (modulo p). This sheds light, since it now places Fermat's Little Theorem in a larger context—there is nothing very special about the integers; for any finite group G , we have

$$x^{|G|} = e$$

for all $a \in G$. And we can prove this in exactly the same way we proved Fermat's Little Theorem, or we can make use of higher-tech group theory ideas such as Lagrange's Theorem.¹

Yet something seems missing. We are happy to see Fermat's Little Theorem sitting neatly inside the larger context of group theory, but the original question was purely number-theoretical, and moreover, it asserted $a^p \equiv a \pmod{p}$ for any integer a . The group-theoretic argument leaves out the "trivial" case. Is there a way to prove both cases at the same time?

Once again, we shall appeal to combinatorics. Following the previous example, we ask whether

$$\frac{a^p - a}{p}$$

counts something simple. And indeed it does: Consider a necklace made with p beads, where each bead can be any one of a colors. How many different necklaces are there? Certainly, if one necklace can be "rotated" into another, then they are the same necklace. For example, let $p = 5$. If one necklace reads blue-blue-blue-red-red clockwise starting from some arbitrary bead, it is the same as a necklace that reads red-blue-blue-blue-red. There are ap different strings, and the number of different necklaces should be equal to this number, divided by p .

Not quite! This assumes that counting different strings is uniformly overcounting by a factor of p . If the string is monochrome, for example, red-red-red-red-red, the 5 rotations all produce the same string. On the other hand, as long as a string is not monochrome, then the 5 rotations *will* produce 5 different strings. This is a direct consequence of the primality of 5. Strings with, say, 6 beads do not have this non-monochrome-uniform-overcounting property (look at red-red-red-blue-blue-blue).

Now we are ready to deduce Fermat's Little Theorem: There are a monochrome strings, so there are $a^p - a$ different non-monochrome strings. Thus there are

$$\frac{a^p - a}{p}$$

non-monochrome necklaces, an integer, and we are done.

Fermat's Little Theorem thus has a deep (group theory) and elementary (beadcounting) meaning. Seeing it from both of these viewpoints enriches our understanding of this not-so-little theorem, and helps us to understand *why* it is true.

3 Complex Numbers

Our goal is to show *why* Euler's formula is true, but first we need to look at something more fundamental:

Why do complex numbers multiply so nicely?

In other words, if $|z| = r$, $\arg(z) = \alpha$ and $|w| = s$, $\arg(w) = \theta$, show *why*

$$|zw| = rs \quad \text{and} \quad \arg(zw) = \alpha + \theta.$$

It is easy to see *how* this is true. Just write

¹ If k is the smallest positive integer such that $ak = e$, then $\{a, a^2, \dots, a^{k-1}, ak = e\}$ will be a k -element subgroup of G . By Lagrange's Theorem, k must divide $|G|$. Thus $a^{|G|} = a^{km} = (a^k)^m$ for some integer m .

$$z = r(\cos \alpha + i \sin \alpha), w = s(\cos \theta + i \sin \theta),$$

and let trig identities do the rest. But that teaches you nothing; the trig functions are merely algebraic placeholders. And of course, using Euler's Formula is cheating! Instead, change the point of view from algebra/trig to physics. Think of complex numbers as *vectors* in the plane, and focus on the only essential multiplication that needs to be understood, namely:

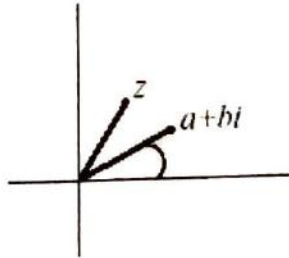
What happens when the complex number z is multiplied by i ?

We need only check two cases: When a real number r is multiplied by i , the result is ir . When the purely imaginary number it is multiplied by i , we get $-t$. In both cases, multiplication by i is equivalent to rotation of 90 degrees counterclockwise. Since multiplication by i does the same thing to both components of the complex vector, we have answered our question, and determined that

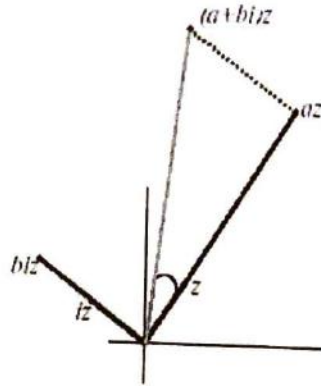
zi is the vector z , rotated 90 degrees counterclockwise, for any complex number z .

Now we are ready to get to work. Imagine multiplying the complex number z by $w = a + bi$. We will concede only the tiniest algebra:

$$zw = az + ibz.$$



Now *visualize* this, thinking vectorially: Start at the origin, and move to az , a complex number with the same angle as z , but with magnitude $a|z|$. Our current $\arg$ is α . Now we must add the complex number ibz . Thus we now turn 90 degrees counterclockwise, and move $b|z|$ units. The resulting path travelled was the legs of a right triangle with lengths in $a|z|$, $b|z|$. The magnitude is thus clearly $|a + bi||z| = |w||z|$, and since the legs are in the ratio $a : b$, the angle traveled is $\arg(a + bi) = \arg(w) = \vartheta$, which must be added to the initial angle of α . So the final angle is $\alpha + \vartheta$.



This wonderful demonstration, due to Tristan Needham² truly shows us *why* complex numbers multiply the way they do. And, if you care to think about it, this demonstration also gives you good proofs of the trig identities that the old How argument needed!

Why is Euler's Formula True?

All investigations of Euler's formula must face the question of extending imaginary numbers to the domain of the exponential function. The power-series proof assumed that it is "legal" to plug imaginary numbers into the real power series. In other words, the power series for the exponential function can be thought of as its *defining property*. But another defining property of exponentials is the unique differentiation rule, namely

$$\frac{d}{dt}(e^{at}) = ae^{at}.$$

for all real numbers a . Now, extend this to imaginary numbers, and we have

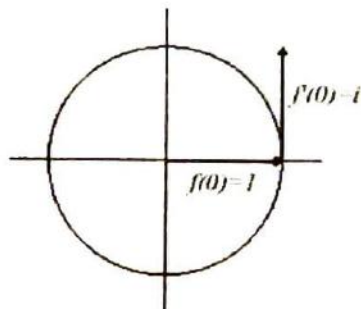
$$\frac{d}{dt}(e^{it}) = ie^{it}.$$

Let us think like a physicist again. Define $f(t) = e^{it}$ to be some as-yet-unknown vector function of time t . We know that $f(0) = 1$, and we also know that $f'(t) = if(t)$. Can we solve this differential equation? Easily, once we recall the meaning of multiplication by i : the instantaneous velocity vector of $f(t)$ has the same magnitude as $f(t)$ and is pointing 90 degrees counterclockwise to $f(t)$. This is describing the physics of circular motion! In other words, a circle (traveling counterclockwise at unit speed) satisfies these dynamic properties, and since solutions of differential equations are unique, we are done. We can stop by saying that $ei t$ means counterclockwise circular motion at unit speed, starting at 1, but if we insist on symbolic notation, we can say

² From Needham's *Visual Complex Analysis*, (Clarendon, 1997). This book is also our source for the "physics" demonstration of Euler's Formula. Indeed, this book contains a fantastic wealth of Whys concerning complex numbers, geometry, and linear algebra.

$$e^{it} = \cos t + i \sin t.$$

but notice that this has no more content than saying “circular motion.”



In other words, the physics is the fundamental point of view. Algebra and trigonometry are mere middlemen; placeholders. And if we want to explain both complex multiplication and Euler's formula simultaneously with one concept, it would not be oversimplifying much to state that the reason *why* these are true is due to the “rotate by 90 degrees counterclockwise” meaning of i .

4 An Algorithmic Point of View

Let us turn to the Arithmetic-Geometric Mean Inequality (AGM), which states that for positive $x_1, x_2, \dots, x_n$,

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}.$$

with equality if and only if $x_1 = x_2 = \dots = x_n$. Why is it true?

There are many interesting proofs for this. One of the neatest How proofs, due to Cauchy, uses induction to prove it for $n = 2r$ (the case $n = 2$ can be proven with brute force, easily). Then Cauchy used a clever substitution to reduce n *downwards* to fill in the gaps. For example, suppose we know it is true for $n = 4$. We get $n = 3$ using the four variables

$$a, b, c, \frac{a + b + c}{3}.$$

Then the $n = 4$ case implies

$$\frac{a + b + c + (a + b + c)/3}{4} \geq \sqrt[4]{abc \left(\frac{a + b + c}{3} \right)}.$$

Let $A = (a + b + c)/3$. Then we have

$$A \geq \sqrt[4]{abcA}.$$

Thus

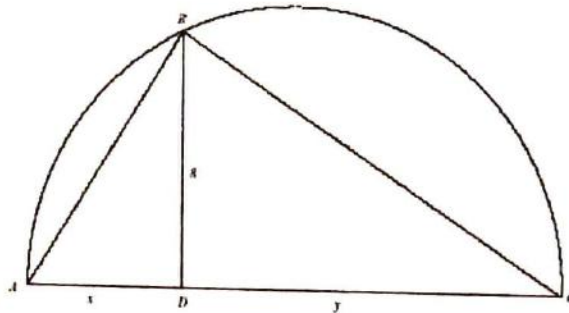
$$A^4 \geq abcA,$$

so

$$A \geq \sqrt[4]{abc}.$$

and we're done.

This is wonderful, magical algebra, but why did it work? It is not too hard to see, if we look at things from the right point of view. Let's avoid algebra for the time being, and first prove the $n = 2$ case with a picture of a right triangle ABC inscribed in a semicircle. The altitude BD has length g , and $AD = x$ and $DC = y$. Thus the diameter of the semicircle is $(x + y)$.



Imagine sliding point D along line AC . Clearly, BD achieves maximum when D is at the midpoint of AC , in which case $BD = (x + y)/2$. By similar triangles, $g/x = y/g$, so $g = \sqrt{xy}$. Since $g = BD$, we have effortlessly proven that

$$\frac{x + y}{2} \geq \sqrt{xy},$$

the $n = 2$ case of AGM.

But we've proven much more, because of the dynamic nature of our argument. As D slides from left to right, we see that BD increases from zero steadily until D is at the midpoint, and then it decreases (by symmetry) back to zero. This is equivalent to the intuitively obvious *Symmetry-Product Principle*:

As the distance between two positive numbers decreases, their product increases, provided that their sum stays constant.

We shall now use this principle to prove AGM, with physical, algorithmic point of view. First we will reformulate it, to make it more like the symmetry-product principle.

AGM Reformulated

Let's visualize the positive real numbers $x_1, x_2, \dots, x_n$ as points on the number line, with sum $S = x_1 + x_2 + \dots + x_n$ and product $P = x_1 x_2 \dots x_n$. Then AGM says that

$$\frac{S}{n} \geq \sqrt[n]{P},$$

which is equivalent to

$$P \leq \left(\frac{S}{n}\right)^n.$$

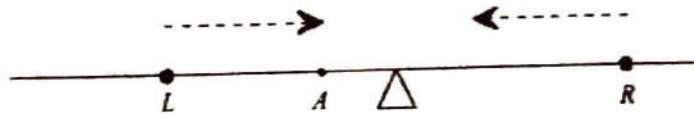
In other words, AGM is equivalent to

If S is held constant, then the largest value of P is attained when all the x_i are equal, i.e., when

$$x_1 = x_2 = \cdots = x_n = \frac{S}{n}.$$

Let us prove this. Note that $A := S/n$ can be interpreted as the “balancing point” of the n points. So start with $x_1, x_2, \dots, x_n$, balancing at A . We shall try to maximize the product P by moving the x_i around, but leaving S (and hence also A) unchanged. Here’s a physically intuitive algorithm:

1. If all the points are already equal to A , stop.
2. If not, then at least one point, L , will be to the left of A , and another one, R , will be to the right of A . Select the one that is closest to A , and move it to A . In order to keep the sum S constant, move the other one accordingly. For example, in the picture below, L was closer to A , so it moves to A , while R moves an equal distance to the left (but ends up to the right of A).



The fulcrum in the picture represents the balancing point for just L and R . Since L and R moved equal distances, this fulcrum stayed unchanged. In other words, the sum of L and R did not change. However, the distance between them did change; it decreased. So by the symmetry-product principle, applied to these two points, the product of L and R increased.

Notice that changing the two points L and R did not effect the other $n - 2$ points, nor did it change their sum. But it did increase the product P . If we keep applying step #2, we will keep increasing the product P while keeping S constant. But eventually we will be unable to make anymore changes, since in a finite number of steps all of the values will equal A . Thus, eventually, the product will be maximized at An . We’ve proven AGM, with virtually no algebra! The physical-algorithmic point of view taught us *why* AGM is true.

5 Probability

Our next example, unlike the previous ones, is not a famous theorem. It is a puzzle that the author used as a (very hard) math contest question in 2000:

Consider the following experiment:

- First a random number p between 0 and 1 is chosen by spinning an arrow around a dial which is marked from 0 to 1. (This way,

the random number is “uniformly distributed”—the chance that p lies in the interval, say, from 0.45 to 0.46 is exactly $1/100$; and the chance that p lies in the interval from 0.324 to 0.335 is exactly $11/1000$, etc.)

- Then an unfair coin is built so that it lands “heads up” with probability p .

- This coin is then flipped 2000 times, and the number of heads seen is recorded.

What is the probability that exactly 1000 heads were recorded?

The probability of seeing k heads in n tosses is easily seen to be

$$U(n, k) := \int_0^1 \binom{n}{k} p^k (1-p)^{n-k} dp.$$

This is a nasty integral, so rather than trying to figure out $U(2000, 1000)$ from scratch, it is worthwhile to check small values of n and k . Experimentation leads to the intriguing conjecture that for any fixed n ,

$$U(n, k) = \frac{1}{n+1}$$

for *all* values of k ! Thus, the answer to the problem appears to be $1/2001$, and this is also the probability of seeing 0 heads, or 2000 heads, or 1345 heads, etc. In other words, the probabilities appear to be uniformly distributed!

One can, with ingenuity, crank out the integral. Several bright high school students did just that in 2000, one by repeated application of integration by parts, and another by cleverly manipulating binomial series. So indeed, the answer is $1/(n+1)$.

But these methods are messy, and they only show us how, not why.

Generating Function Solution to the Coin Problem

A colleague of the author's, when challenged to find out *why* the answer is $1/(n+1)$, came up with the following elegant proof, using the method of generating functions. Fix n , and define $u_k := U(n, k)$. Now look at

$$f(x) := u_0 + u_1 x + u_2 x^2 + \cdots + u_n x^n.$$

Substituting the integral formula for u_k , we have

$$f(x) = \sum_{k=0}^n \left(\int_0^1 \binom{n}{k} p^k (1-p)^{n-k} dp \right) x^k.$$

Interchanging the summation and integration, we get

$$\begin{aligned} f(x) &= \int_0^1 \left(\sum_{k=0}^n \binom{n}{k} p^k x^k (1-p)^{n-k} \right) dp \\ &= \int_0^1 (1-p+px)^n dp \quad (\text{by the Binomial Theorem}) \end{aligned}$$

$$\begin{aligned}
&= \left. \frac{(1-p+px)^{n+1}}{(n+1)(x-1)} \right|_0^1 \\
&= \frac{1}{n+1} \left(\frac{x^{n+1}-1}{x-1} \right) \\
&= \frac{1}{n+1} (x^n + x^{n-1} + \cdots + x + 1).
\end{aligned}$$

In other words, all the coefficients uk are equal to $1/(n+1)$. My colleague triumphantly declared, "That's why!"

Algorithmic Solution to the Coin Problem

But no, it isn't why. The generating function proof is beautiful, but magical. The punchline creeps up without warning. It entertains, but does not enlighten.

Here's *why* the probability distribution is uniform. Again, we will use an algorithmic point of view. Imagine simulating 4 tosses on a computer, for example, with a spreadsheet. Cell A2 below contains a uniformly distributed random number; that is the value for p . To simulate the 4 tosses, we generate 4 more random numbers, and check to see if they are greater than or less than p , i.e., greater than or less than the value in A2. If they are less than p , we output "H," otherwise, we output "T." This clearly accomplishes the simulation.

	A	B	C	D	E
1	p	toss 1	toss 2	toss 3	toss 4
2	0.8424	0.4426	0.8563	0.4192	0.8407
3		H	T	H	H

How many heads will we see? If all the cells in the range B2:E2 were greater than A2, there would be zero heads. On the other hand, if they were all smaller than A2, there would be 4 heads. If, as in the picture, only one is greater than A2, then there are 3 heads. In other words,

The number of heads depends purely on the relative ranking of A2 among the 5 random numbers.

But since these numbers are all uniformly distributed, the rank of any one number is also uniformly distributed; i.e., it is just as likely that A2 is the largest as it is the smallest as it is the 3rd largest, etc. So the probabilities for all 5 outcomes are the same, namely, $1/5$.

Integration and generating functionology obscured a relatively simple situation. The probabilities were uniform because the numbers were uniform, and thus their rankings were uniform.

6 Counting "Cube Root of Unity" Permutations

We shall conclude with a plea for help, with a problem whose solution—at least for the author—has no *Why*, but only *How*. Like the previous example, it appeared in a math contest, the Bay Area Math Olympiad, in 2001.

For each positive integer n , let a_n be the number of permutations τ of $\{1, 2, \dots, n\}$ such that $\tau(\tau(\tau(x))) = x$ for $x = 1, 2, \dots, n$. The first few values are

$$a_1 = 1, a_2 = 1, a_3 = 3, a_4 = 9.$$

Prove that 3^{34} divides a_{2001} .

Consider the permutations τ of $\{1, 2, \dots, n\}$ such that $\tau^3 = \text{id}$. Then for each $x \in \{1, 2, \dots, n\}$, there are only two possibilities:

- Either x is a **fixed point**; i.e., $\tau(x) = x$,
- Or x is a member of a **3-cycle** (xyz) .

We can thus partition these permutations into two cases:

- Either 1 is a fixed point, or it is not. In the first case, the remaining elements $\{2, \dots, n\}$ can be permuted in a_{n-1} ways.
- Or 1 is part of a 3-cycle $(1ij)$, where $i \neq j$ and $i, j \in \{2, \dots, n\}$. There are $(n-1)(n-2)$ such 3-cycles (order counts). Then the remaining $n-3$ elements can be permuted in a_{n-3} ways.

Hence we have (for $n > 3$)

$$a_n = a_{n-1} + (n-1)(n-2)a_{n-3}. \quad (1)$$

Numerical Experimentation

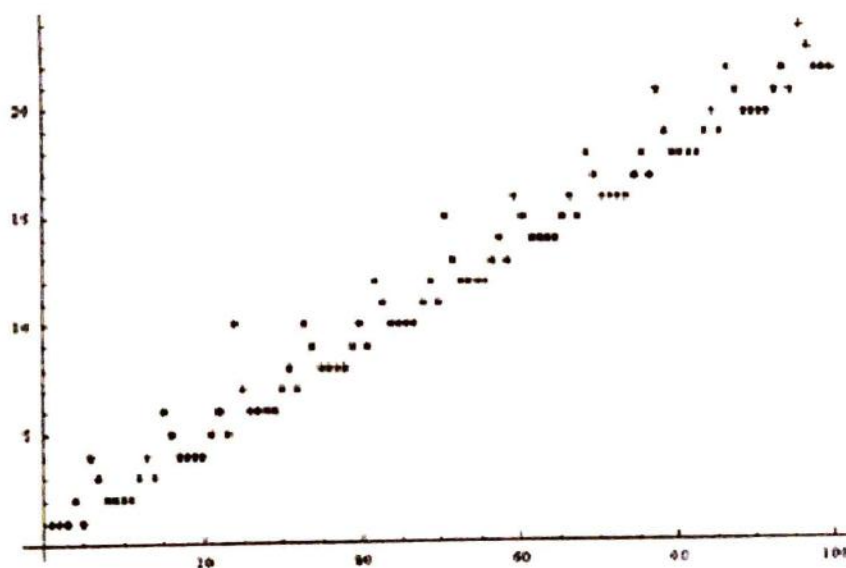
Armed with the recurrence relation (1), it is easy to compute values of a_n . The first few values are

$$1, 1, 3, 9, 21, 81, 351, 1233, 5769.$$

It is striking that all the values after the first two, are divisible by 3. In fact, the powers of 3 that divide the a_n seem to increase quickly. Define b_n such that $3^{b_n} \mid a_n$, i.e., b_n is the highest power of 3 which divides a_n . Here is a table.

n	b_n
1	1
2	1
3	1
4	2
5	1
6	4
7	3
8	2
9	2
10	2
11	2
12	3
13	4
14	3
15	6
16	5
17	4
18	4
19	4
20	4

Indeed, the exponent of 3 seems to rise almost monotonically. Is there a pattern? A graph of the first hundred values of b_n shows a striking regularity that just cannot be a coincidence.



The graph suggests that b_n is approximately $n/4$. We offer a proof of a weaker statement, that essentially $b_n \geq n/6$:

If $n \not\equiv 0 \pmod{3}$, then $(n-1)(n-2)$ will be a multiple of 3, so (1) yields

$$a_n = a_{n-1} + 3ka_{n-3}$$

where k is a positive integer. Hence

$$b_n \geq \min(b_{n-1}, b_{n-3} + 1), \quad n \not\equiv 0 \pmod{3}. \quad (2)$$

If $n \equiv 0 \pmod{3}$, then $(n-1)(n-2) \equiv -1 \pmod{3}$, but both $(n-2)(n-3)$ and $(n-3)(n-4)$ are multiples of 3. Plugging into (1), we have

$$\begin{aligned} a_n - a_{n-1} &= (n-1)(n-2)a_{n-3} = (3u-1)a_{n-3}, \\ a_{n-1} - a_{n-2} &= (n-2)(n-3)a_{n-4} = 3va_{n-4}, \\ a_{n-2} - a_{n-3} &= (n-3)(n-4)a_{n-5} = 3wa_{n-5}, \end{aligned}$$

for positive integers u, v, w . Thus

$$a_n - a_{n-3} = (3u-1)a_{n-3} + 3va_{n-4} + 3wa_{n-5},$$

so

$$a_n = 3(ua_{n-3} + va_{n-4} + wa_{n-5}).$$

Hence

$$b_n \geq 1 + \min(b_{n-3}, b_{n-4}, b_{n-5}), \quad n \equiv 0 \pmod{3}, n > 5. \quad (3)$$

Applying (2) and (3) repeatedly yields

$$b_{6k+3} \geq k + 1.$$

The role played by 3 could be played by any prime, where we would use Wilson's Theorem, $(p-1)! \equiv -1 \pmod{p}$, in the line after (2). This is a nice generalization, but we are, of course, not satisfied at all. Not only have we failed to prove the stronger result $bn \approx n/4$, the argument that we did find is just—algebra. There do seem to be stronger arguments, but they use generating functions and p -adic convergence, intriguing, but not very natural.³

Yet the combinatorial definition of a_n —the number of permutations on n objects whose cube is the identity—demands that one think about powers of three. There *should* be a neat combinatorial argument, partitioning the permutations in some clever way, along the same lines of the combinatorial proof of Fermat's Little Theorem, only more intricate. Unfortunately, we have failed in all of our attempts to find such an argument. So we end with the simple question, "Why?"

The answer is up to you, the reader.

Acknowledgements

The author is indebted to Bjorn Poonen³, John Kao, and especially, to Doug Jungreis and Tristan Needham for many valuable discussions.

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³ Bjorn Poonen, personal communication.

HCSSiM
Hampshire Summer Studies in Mathematics
Betty Luan
Student of HCSSiM 2001

Every summer, twenty-five highly motivated secondary school students come across the United States and Canada to gather at the rigorous six-week math program in Amherst, Massachusetts.

Located on the rural side of Massachusetts, the 550-acre Hampshire College campus is surrounded by farmlands and hills of the Holyoke Range. The tranquil and pleasant atmosphere sets the environment for creativity, enjoyment, and interaction. On some occasions, campers hike the Holyoke Range, climbing over many peaks to their destination for a picnic or a sunrise hike above the wonderful vista.

In small classes, students engage in intense learning. Topics such as topology, set theory, number theory, graph theory, probability, theoretical computer programming, combinatorics, complex numbers, four-dimensional geometry, and metric spaces involve demanding mathematical thought, opening up many areas of mathematics to math lovers.

From Monday to Saturday, students attend a four-hour morning class taught by university professors and talented graduate and undergraduate students. Rather than teaching the results of math discoveries, students are taught the processes and methods to reach those discoveries.

In the afternoon, students attend a one-hour Prime Time session, when a special guest provides a fulfilling one-hour lecture on a math topic such as Fermat's Last Theorem, Wallpaper Design, Minkowski's Theorem, Hyperplanes and Ergodic Theory.

In the evening, students participate in a challenging and exciting three-hour problem-solving session. Often you will hear people jumping up from their chairs to rejoice as they finally solved a problem after hours of brain-racking thought.

HCSSiM can be a fulfilling experience in both academic learning and social interaction. In the afternoon between classes, campers might play a crazy game of ultimate or visit the recreation center where they have access to pools, ping-pong tables, basketball courts, and the weight room. Winding through Hampshire campus is a mile long road, flanked by woodlands, for jogging and biking in the grass-fresh atmosphere.

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As an alumna of the HCSSiM 2001 program, I can say that it has been my most memorable experience. In the evening, from 7:30 until 10:30 during problem solving session, we worked in groups when a problem required cumulative brain power; otherwise, we worked individually. Many nights I struggled with just one problem. I felt that I knew how to solve the problem, so I kept at the problem for a few hours. Sometimes, everything became clear and I jumped up from my chair to scream "Oh Yeah!" Other times, I wasn't able to solve them. One of my Canadian friends, Max, drew on an orange with a marker to solve a problem involving spherical triangles. I think he ate it later.

Most memorable about the program are the people. I've had some of my best times with them during the trip to Boston, the Sunday outing at Tanglewood Symphony, the five-hour hikes, and our all-nighter Saturday talks. The teachers also have a great sense of humor. My most meaningful experience that I wrote for my SSR questionnaire, without question, was about HCSSiM - the sunrise hike.

It was the last night of the six week Hampshire College summer math camp when the twenty-five of us spent half the night talking and writing parting messages. Around three thirty in the morning, everyone changed into comfortable clothes and prepared for the sunrise hike. With only two flashlights, we walked across campus and soon entered a field of tall weeds. We pushed through the weeds with one person in front of the single-file line carrying the flashlight and another person in the middle carrying the other. It was so dark that the flashlights didn't help much. It was by word of mouth, passed along from one to the other down the line, that everyone knew what to expect on the ground - a log, a ditch, or possibly a rock.

We soon entered a dark forest. The trees swayed above us but we couldn't see them. We also knew that the path was narrow and on both sides of the path were steep drops that fell into a dried lake. We followed the steps of the person ahead of us, trusting that person. We rounded a few turns uphill and swung the flashlights wildly across our path.

An hour or so later, dark purple color streaked across the sky. Climbing faster, we tried to race with the rising sun. Soon, layers of delicate violet strokes appeared on the dawning sky, revealing silhouettes of outreaching branches on either side as we rushed on. It was getting lighter, and it became easier to see our way. We turned off the flashlights and ran up the rest of the way to the top. We stumbled onto the rocky summit and peered over the cliff, just in time to catch a glimpse of the bright sphere slipping above the horizon.

That morning, we sat silently along the edge of the cliff watching the vast land below us and the brightening sun. No one spoke, but everyone knew we would disperse in a couple of hours, possibly not to be together for quite a while. We watched the sky, the land, the sun, marking the place with our thoughts.

It was a strange hour. Everyone was silent and gazing peacefully at the vast rural land below us. The hike gave me a deep appreciation for life and nature. Though I was above everything and was able to see very far, I felt smaller than ever.

"I wish this program wouldn't stop." (Joseph Ahn, Massachusetts)

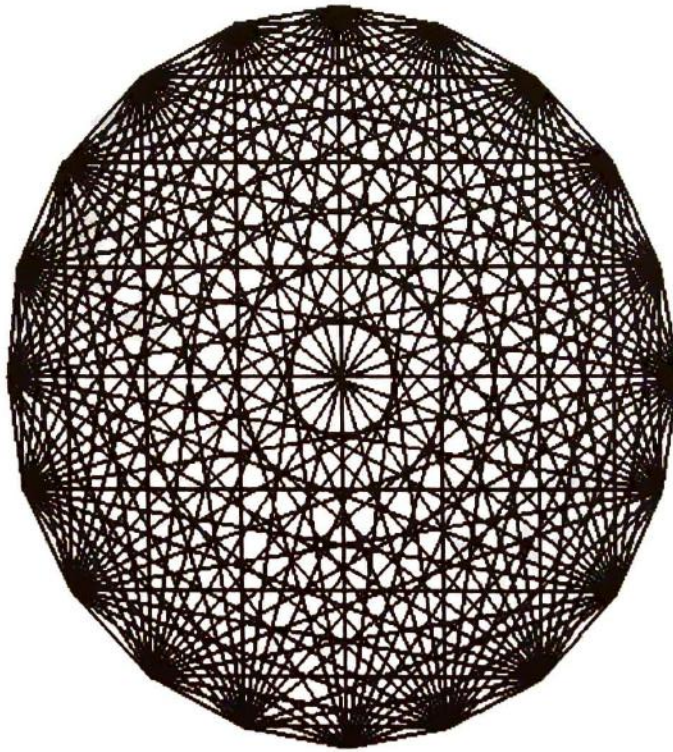
"In one week at HCSSiM, I have learned more math than I did in 2 years in High School." (David Levitt, New York)

"There is NO such thing as a boring moment!!!" (Tim Lewis, Pennsylvania)

You can visit the HCSSiM website at www.hcssim.org.

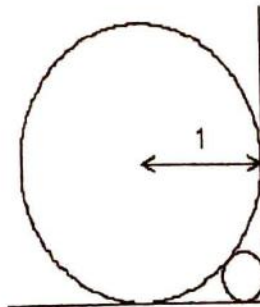
Problems

1. A Mystic Rose consists of twenty points evenly spaced around a circle and every point is joined to every other point.



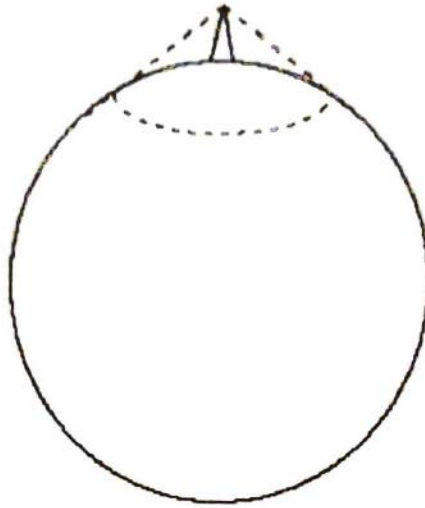
How many lines are required to construct the Mystic Rose?

2. A unit circle is placed against a right angle.



What is the radius of the smaller circle?

3. From how far on a clear day, and assuming there are no obstructions, could you see from the top of the Eiffel tower?



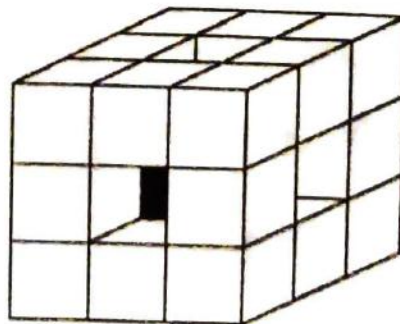
Take the height of the Eiffel tower as 300 meters and the radius of the Earth as 6240 km.

4. If the number 41 is added to its reverse, $41 + 14 = 55$.

The total, 55, is divisible by 5. But not many numbers have this property. For example, $27 + 72 = 99$, which is not divisible by 5.

How many numbers under 100 have this property?

5. A girl bought 15 pens costing £1.84. She paid one pence more for each red pen than each blue pen. How many of each kind did she buy and at what price?
6. A cube measuring $3 \times 3 \times 3$ is made up of 27 smaller cubes and has a 1×1 square hole pushed right through the center of each face so that you can see straight through the cube from every side.



The number of small cubes remaining is 20.

If a $5 \times 5 \times 5$ cube has 3×3 square holes pushed through the centre of each face, how many smaller cubes would remain?

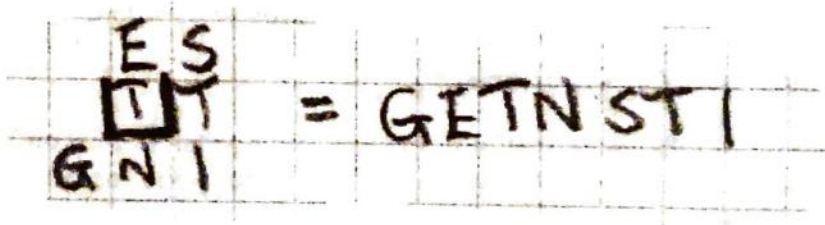
7. In order to win a game of darts, a player must finish on exactly zero and their last dart must land in a double or hit the bull's eye.

For example, if a player hit treble twenty (60 points), double twenty (40 points) and the bull's eye (50 points) they would score 150 points and could use this combination to finish.

How many ways can a player finish from 150 points with three darts?

8. During the first lesson with your new and rather eccentric mathematics teacher he continues his tradition of telling the class that if anyone can guess his favorite mathematician they will be let off homework for the rest of the year. However, if you make a wrong guess you are duty bound to wash his car every week for the rest of the year. In all the years he has taught, the sparkle on his car has been the envy of every other teacher. Despite this seemingly impossible challenge, one person has managed to work it out... your older brother!

Of course he is not going to tell you the answer, but after a little 'research' you find two scraps of paper hidden inside one of his books. The first piece seems to hold a clue.

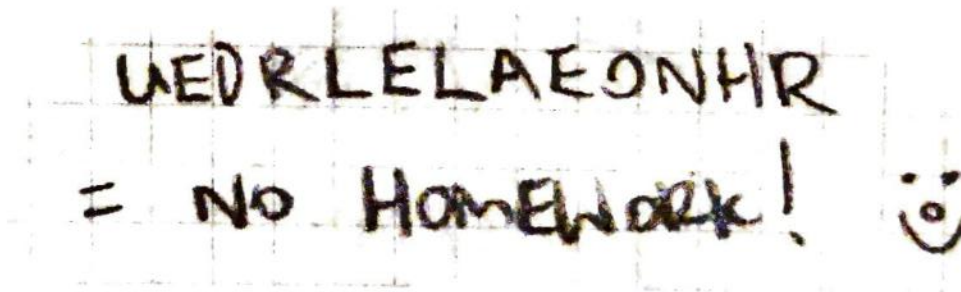


Handwritten clue on a grid:

E	S						
I	T						
G	N						

= GETNSTI

The second piece is obviously the answer you are seeking.



Handwritten answer on a grid:

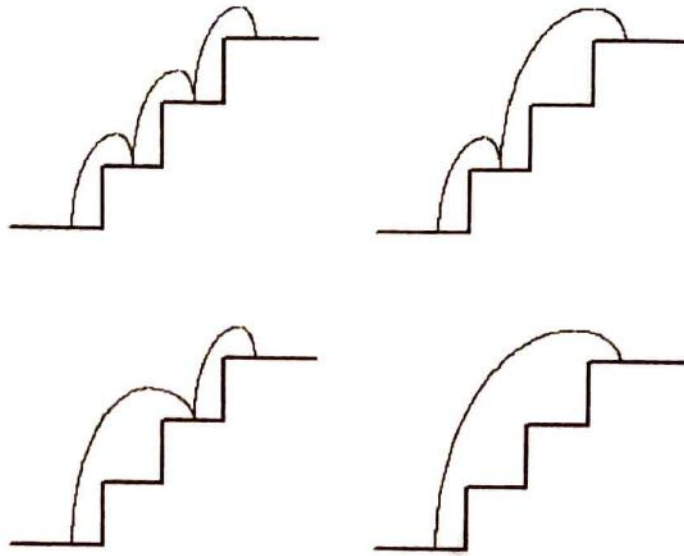
U	E	D	R	L	E	L	A	E	N	H	R
---	---	---	---	---	---	---	---	---	---	---	---

= NO HOMEWORK! :)

9. The product of the digits in the number 126, is $1 \times 2 \times 6 = 12$.

How many other three digit numbers have a product that is equal to 12?

10. It is possible to climb three steps in exactly four different ways.



How many ways can you climb ten steps?

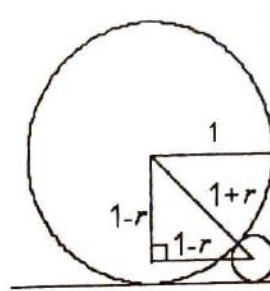
Solutions

1. Starting at the first point, it must be joined to nineteen other points. The second point only needs to be joined to eighteen points now. The third point needs to join seventeen points and so on.



Number of lines, $L = 19 + 18 + 17 + \dots + 1 = \frac{1}{2} \times 19 \times (19 + 1) = 190$.

2. Let the radius of the small circle be r .



Using Pythagoras' Theorem, $(1 + r)^2 = (1 - r)^2 + (1 - r)^2 = 2(1 - r)^2$

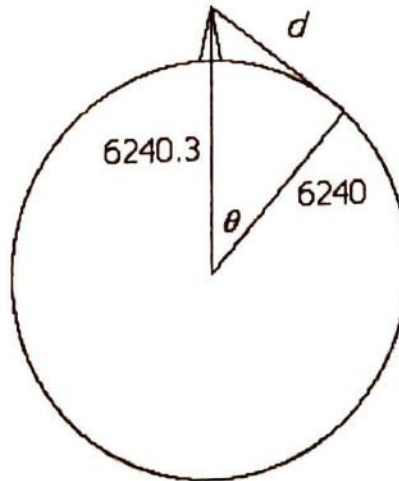
$$1 + 2r + r^2 = 2(1 - 2r + r^2) = 2 - 4r + 2r^2$$

$$r^2 - 6r + 1 = 0$$

Solving the quadratic we get $r = 3 \pm 2\sqrt{2}$, giving the solution $r = 3 - 2\sqrt{2}$.

3. We shall answer this question in two ways:

Let us begin by considering the distance from the top of the tower to furthest point on the surface of the earth (the tangential length).



Applying the Pythagorean Theorem, $6240.3^2 = d^2 + 6240^2$. Hence $d \approx 61.2$ km.

However, this does not represent the distance along the curvature of the earth (the arc length). To do this we apply trigonometry to the right angle triangle above, $\cos \theta = 6240/6240.3$, thus $\theta \approx 0.562$ degrees.

As the radius of the earth is 6240 km, the circumference is $2\pi \times 6240 \approx 39187$ km. So the arc length will be $0.562/360 \times 39187 \approx 61.2$ km, which is the same answer as the other method... coincidence?

4. Let the original number, $n = 10a + b$, so its reverse, $m = 10b + a$.

Therefore, $n + m = 11a + 11b = 11(a+b)$.

Clearly, $n + m$ is divisible by 5 if $a + b \equiv 0 \pmod{5}$.

Listing combinations:

$a = 0, b = 5$	05, 50
$a = 1, b = 4$ or 9	14, 41, 19, 91
$a = 2, b = 3$ or 8	23, 32, 28, 82
$a = 3, b = 7$	37, 73
$a = 4, b = 6$	46, 64
$a = 5, b = 5$	55
$a = 6, b = 9$	69, 96
$a = 7, b = 8$	78, 87

That is, 19 numbers in total.

5. Let a be the number of blue pens, b be the number of red pens and c the cost of a blue pen, hence a red pen costs $c + 1$.

$$\rightarrow ac + b(c + 1) = 184$$

$$\rightarrow ac + bc + b = 184$$

$$\rightarrow ac + bc = 184 - b$$

$$\rightarrow c(a + b) = 184 - b$$

$$\text{But } a + b = 15,$$

$$\rightarrow 15c = 184 - b$$

Hence $184 - b$ must be divisible by 15; as $15 \times 12 = 180$, we get $b = 4$, so $a = 11$ and $c = 12$.

i.e. 11 blue pens at 12p each and 4 red pens at 13p each, costing £1.84 in total.

6. A solid cube measuring $5 \times 5 \times 5$ consists of 125 cubes.

A cube measuring $3 \times 3 \times 3 = 27$ cubes is removed from the center and six faces of $3 \times 3 = 9$ cubes are removed.

That is, $125 - (27 + 6 \times 9) = 125 - (27 + 54) = 125 - 81 = 44$ cubes remaining.

7. If we finish on a bull's eye: $150 - 50 = 100$, we must make 100 with the first two darts: 40,60,50 / 50,50,50 / 60,40,50

The highest double to finish on is $2 \times 20 = 40$ and $150 - 40 = 110$; which can be made in two ways: 50,60,40 / 60,50,40

The highest score possible with first two darts is $2 \times 60 = 120$, so the lowest double is $150 - 120 = 30$, i.e. double 15.

With the exception of finishing on a bull's eye the first two darts must score over 100, which means they must both be trebles and so the sum of their scores will be divisible by 3. As 150 is divisible by 3, the double must also be divisible by 3.

Double 15, $150 - 30 = 120$: 60,60,30

Double 18, $150 - 36 = 114$: 54,60,36 / 60,54,36 / 57,57,36

And so there are 9 ways of finishing on 150.

8. The encryption process is simple. Starting inside the center box, the plain text is written up and outward in a clockwise spiral. Then the cipher text is formed by reading down the columns.

The secret of decryption is to produce a spiral of numbers, starting at 1 and counting up to the number of letters in the cipher text. In our cipher, the number of letters is 13:

1	1	1	1
0	1	2	3
9	2	3	
8	1	4	
7	6	5	

This provides a map to overlay the cipher, UEDRLELAEONHR, and it is then a simple matter of writing it down the columns:

U	L	E	R
E	E	O	
D	L	N	
R	A	H	

Hence the name we seek is, of course, *Leonhard Euler*.

9. The three digits must be (126), (134) or (223) and there are fifteen different 3-digit numbers made up of these digits:

126, 162, 216, 261, 612, 621, 134, 143, 314, 341, 413, 431, 223, 232, 322.

10. If there are ten steps, it is necessary to arrive at the 10th step, but all the other steps in between are optional.

Step Number	1	2	3	4	5	6	7	8	9	10
	0	0	0	1	0	1	0	0	0	1

So the binary sequence, 0001010001, represents a journey using steps 4, 6 and 10. To find the total number of ways of climbing 10 steps we need to consider how many binary sequences can be made using 9 binary digits (as step 10 must always be a 1). Hence there are $2^9 = 512$ ways of climbing ten steps.

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